# Regression Analysis

Lecture notes, Chapters 3–16

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# Chapter 3. Simple Linear Regression

## 3.1 Regression and the straight-line model

Regression analysis studies how a **response** (dependent) variable $Y$ changes with one or more **explanatory** (independent) variables. The name goes back to Galton, who observed that sons of unusually tall or short fathers tend to be closer to the average height: their heights "regress" towards the mean. In Pearson's father–son data the fitted line has slope about 0.5, so a father 6 inches above average has sons only about 3 inches above average on average.

With one explanatory variable and a linear relationship we speak of **simple linear regression**.

**Definition (Simple linear regression model)**

For data $(x_1, y_1), \ldots, (x_n, y_n)$,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2), \qquad i = 1, \ldots, n,$$

where the $x_i$ are treated as fixed (nonrandom) constants and $\beta_0, \beta_1, \sigma^2$ are unknown.

The model packs three assumptions.

(1) **Linearity**: $E(Y \mid X = x) = \beta_0 + \beta_1 x$.

(2) **Normality and equal variance**: given $X = x$, $Y \sim N(\beta_0 + \beta_1 x, \sigma^2)$, with $\sigma^2$ not depending on $x$ (**homoscedasticity**); otherwise the errors are **heteroscedastic** (Chapter 12).

(3) **Independence**: the errors are uncorrelated, $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$ (independent under normality).

The model is linear *in the parameters*; a model such as $\beta_0 + \beta_1 x + \beta_2 x^2$ is still a linear model. If $X$ is itself random, the same analysis applies conditionally on the observed $x_i$; this is the **regression of $Y$ on $X$**, which differs from the regression of $X$ on $Y$.

It is often convenient to centre the explanatory variable:

$$y_i = \beta_0^* + \beta_1 (x_i - \bar{x}) + \epsilon_i, \qquad \beta_0^* = \beta_0 + \beta_1 \bar{x}. \tag{3.1}$$

## 3.2 Least squares

**Definition (Least squares estimator)**

The **least squares estimators** (LSE) $\hat{\beta}_0, \hat{\beta}_1$ minimize

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

Write

$$S_{xx} = \sum (x_i - \bar{x})^2, \qquad S_{yy} = \sum (y_i - \bar{y})^2, \qquad S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}).$$

**Theorem 3.1 (Normal equations)**

If $S_{xx} > 0$, the LSE is the unique solution of the **normal equations**

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i = \sum y_i, \qquad \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2 = \sum x_i y_i,$$

namely

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

**Proof.** Setting the partial derivatives to zero,

$$\frac{\partial S}{\partial \beta_0} = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0, \qquad \frac{\partial S}{\partial \beta_1} = -2\sum x_i(y_i - \beta_0 - \beta_1 x_i) = 0,$$

gives the normal equations. The first gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$; substituting into the second, $\hat{\beta}_1(\sum x_i^2 - n\bar{x}^2) = \sum x_i y_i - n\bar{x}\bar{y}$, i.e. $\hat{\beta}_1 S_{xx} = S_{xy}$. The Hessian

$$J = 2\begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

has $n > 0$ and determinant $4nS_{xx} > 0$, so it is positive definite. $S$ is a convex quadratic, and the stationary point is its unique minimum. ■

In the centred form (3.1) the normal equations decouple: $\hat{\beta}_0^* = \bar{y}$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$, so the fitted line is

$$\hat{y} = \bar{y} + \hat{\beta}_1(x - \bar{x}).$$

**Example 3.1**

Advertising spending $x$ and sales $y$ for six stores:

| $x$ | 2 | 3 | 5 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| $y$ | 5 | 7 | 8 | 11 | 13 | 16 |

Here $\bar{x} = 34/6 = 5.667$, $\bar{y} = 10$, $S_{xx} = 238 - 34^2/6 = 45.33$ and $S_{xy} = 401 - 34 \cdot 60/6 = 61$. So $\hat{\beta}_1 = 61/45.33 = 1.346$, $\hat{\beta}_0 = 10 - 1.346 \cdot 5.667 = 2.37$, and $\hat{y} = 2.37 + 1.346x$.

**Maximum likelihood.** Under normal errors the log-likelihood is

$$l(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} S(\beta_0, \beta_1).$$

For each $\sigma^2$ it is maximized by minimizing $S$, so the MLEs of $\beta_0, \beta_1$ are the LSEs, and $\hat{\sigma}^2_{\text{MLE}} = S(\hat{\beta}_0, \hat{\beta}_1)/n$.

## 3.3 Properties of the fitted line

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $e_i = y_i - \hat{y}_i$.

**Theorem 3.2**

(a) $\sum e_i = 0$; equivalently $\sum y_i = \sum \hat{y}_i$.

(b) $\sum x_i e_i = 0$.

(c) $\sum \hat{y}_i e_i = 0$.

(d) The fitted line passes through $(\bar{x}, \bar{y})$.

(e) $\sum e_i^2$ is the minimum of $S(\beta_0, \beta_1)$.

**Proof.** (a) and (b) are the two normal equations, since $\sum e_i = \sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$ and $\sum x_i e_i = \sum x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$.

(c) $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$.

(d) $\hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$.

(e) is the definition of the LSE. ■

Geometrically, (a)–(c) say that the residual vector is orthogonal to $\mathbf{1}$, to $\mathbf{x}$, and hence to the fitted vector. This is why the residual plot of $e_i$ against $\hat{y}_i$ (or $x_i$) should show no linear trend.

## 3.4 Decomposition of variation

**Theorem 3.3 (Sum of squares decomposition)**

$$\underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE} + \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR},$$

and $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$.

**Proof.** Write $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ and square. The cross term is $2\sum e_i(\hat{y}_i - \bar{y}) = 2\sum \hat{y}_i e_i - 2\bar{y}\sum e_i = 0$ by Theorem 3.2. Since $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$. ■

$SST$ is the **total** sum of squares, $SSE$ the **residual** (error) sum of squares, and $SSR$ the **regression** sum of squares. Their degrees of freedom are $n-1$, $n-2$ and $1$.

**Definition (Coefficient of determination)**

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \in [0, 1].$$

$R^2$ is the proportion of the variation in $y$ explained by the regression. $R^2 = 1$ means a perfect fit; $R^2 = 0$ means $\hat{\beta}_1 = 0$, i.e. the fitted line is horizontal. The standard error of estimate is $s = \sqrt{MSE}$ with $MSE = SSE/(n-2)$; we will see that $MSE$ is unbiased for $\sigma^2$ (Chapter 4).

**Example 3.2**

In Example 3.1, $\sum y_i^2 = 684$, so $SST = S_{yy} = 684 - 600 = 84$. Then $SSR = 61^2/45.33 = 82.08$, $SSE = 84 - 82.08 = 1.92$ and $R^2 = 0.977$: advertising explains about 98% of the variation in sales.

## 3.5 Correlation

For a random pair $(X, Y)$ the **population correlation** is $\rho_{XY} = \mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}\, X \,\mathrm{Var}\, Y}$, and the **sample correlation** is

$$r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \in [-1, 1].$$

**Theorem 3.4**

(a) $R^2 = r_{xy}^2$ in simple linear regression.

(b) $r_{xy} = \hat{\beta}_1 \, s_x/s_y$, where $s_x^2 = S_{xx}/(n-1)$ and $s_y^2 = S_{yy}/(n-1)$; hence $r_{xy}$ has the sign of $\hat{\beta}_1$.

**Proof.** (a) By Theorem 3.3, $R^2 = \frac{SSR}{SST} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2$.

(b) $r_{xy} = \frac{S_{xy}}{S_{xx}}\sqrt{\frac{S_{xx}}{S_{yy}}} = \hat{\beta}_1 \frac{s_x}{s_y}$. ■

$r_{xy}$ measures only *linear* association. For $x = 0, 1, 2, 3, 4$ and $y = x^2 - 4x + 6 = (6, 3, 2, 3, 6)$, $S_{xy} = 0$ and $r_{xy} = 0$, although $y$ is a deterministic function of $x$.

**Remark**

The slope and the correlation answer different questions: $\hat{\beta}_1$ is the expected change in $y$ per unit of $x$, while $r_{xy}$ is unit-free and symmetric in $x$ and $y$. The regression of $x$ on $y$ has slope $S_{xy}/S_{yy}$, and the product of the two slopes is $r_{xy}^2 \leq 1$; the two lines coincide only if $|r_{xy}| = 1$.

# Chapter 4. Inference in Simple Regression

Throughout, $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and fixed $x_i$ with $S_{xx} > 0$.

## 4.1 Sampling distributions of the estimators

Both estimators are linear in $y$:

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \sum_i k_i y_i, \qquad k_i = \frac{x_i - \bar x}{S_{xx}}, \qquad \sum k_i = 0, \quad \sum k_i x_i = 1, \quad \sum k_i^2 = \frac{1}{S_{xx}},$$

(using $S_{xy} = \sum (x_i - \bar x) y_i$), and $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$.

**Theorem 4.1**

(a) $\hat\beta_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$.

(c) $\hat\beta_0 \sim N\Big(\beta_0,\ \sigma^2\big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\big)\Big)$ and $\operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\dfrac{\bar x \sigma^2}{S_{xx}}$.

**Proof.** (a) $\hat\beta_1$ is a linear combination of independent normals, with mean $\sum k_i(\beta_0 + \beta_1 x_i) = \beta_1$ and variance $\sigma^2 \sum k_i^2 = \sigma^2 / S_{xx}$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = \sum_i \frac{1}{n} k_i \sigma^2 = 0$.

(c) $E\hat\beta_0 = (\beta_0 + \beta_1 \bar x) - \bar x \beta_1 = \beta_0$ and, by (b),

$$\operatorname{Var} \hat\beta_0 = \operatorname{Var} \bar y + \bar x^2 \operatorname{Var} \hat\beta_1 = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\bar x \operatorname{Var} \hat\beta_1.$$

■

**Theorem 4.2**

Let $SSE = \sum e_i^2$ and $MSE = SSE/(n-2)$. Then $SSE/\sigma^2 \sim \chi^2(n-2)$, independent of $(\hat\beta_0, \hat\beta_1)$. In particular $E(MSE) = \sigma^2$.

This is the case $p = 1$ of Theorem 6.4; a direct proof uses an orthogonal transformation whose first two rows span $\mathbf{1}$ and $\mathbf{x} - \bar x \mathbf{1}$, exactly as for the normal sample variance. Consequently

$$\frac{\hat\beta_1 - \beta_1}{\sqrt{MSE/S_{xx}}} \sim t(n-2), \qquad \frac{\hat\beta_0 - \beta_0}{\sqrt{MSE(1/n + \bar x^2/S_{xx})}} \sim t(n-2).$$

**Corollary 4.3 (Intervals and tests for the coefficients)**

$100(1-\alpha)\%$ confidence intervals are

$$\beta_1 :\ \hat\beta_1 \pm t_{\alpha/2}(n-2)\sqrt{\frac{MSE}{S_{xx}}}, \qquad \beta_0 :\ \hat\beta_0 \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

The test of $H_0 : \beta_1 = 0$ rejects when $|t_0| > t_{\alpha/2}(n-2)$, $t_0 = \hat\beta_1 / \sqrt{MSE/S_{xx}}$. If $\sigma^2$ is known, replace $MSE$ by $\sigma^2$ and $t$ by $z$.

**Theorem 4.4 (ANOVA $F$ test)**

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$. The statistic $F_0 = MSR/MSE$, $MSR = SSR/1$, satisfies $F_0 = t_0^2$, so the $F$ test with $F_\alpha(1, n-2) = t_{\alpha/2}(n-2)^2$ is the two-sided $t$ test of $\beta_1 = 0$.

**Proof.** $E(SSR) = S_{xx} E\hat\beta_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Also $SSR/MSE = \hat\beta_1^2 S_{xx}/MSE = t_0^2$. ∎

So $E(MSR)/E(MSE) = 1 + \beta_1^2 S_{xx}/\sigma^2$, which is 1 exactly when $\beta_1 = 0$; large $F_0$ is evidence against $H_0$.

## 4.2 Estimating the mean response and predicting

At a point $x$, the mean response $\mu_{y\cdot x} = \beta_0 + \beta_1 x$ is estimated by $\hat y = \hat\beta_0 + \hat\beta_1 x = \bar y + \hat\beta_1(x - \bar x)$.

**Theorem 4.5**

(a) $\hat y \sim N\Big(\mu_{y\cdot x},\ \sigma^2\Big(\dfrac{1}{n} + \dfrac{(x-\bar x)^2}{S_{xx}}\Big)\Big)$, and a $100(1-\alpha)\%$ confidence interval for $\mu_{y\cdot x}$ is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

(b) For a new observation $y_s = \beta_0 + \beta_1 x + \epsilon_s$, independent of the sample, $y_s - \hat y \sim N\big(0, \sigma^2(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}})\big)$, and a $100(1-\alpha)\%$ prediction interval is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

**Proof.** (a) By Theorem 4.1(b), $\operatorname{Var}\hat y = \operatorname{Var}\bar y + (x-\bar x)^2 \operatorname{Var}\hat\beta_1$. Studentize with $MSE$, which is independent of $\hat y$.

(b) $y_s$ is independent of $\hat y$, so the variances add. ∎

Both intervals are narrowest at $x = \bar x$ and widen quadratically away from it; extrapolation is costly. The prediction interval does not shrink to zero as $n \to \infty$, because of the irreducible variance $\sigma^2$ of a new observation.

**Example 4.1**

For the advertising data of Example 3.1 ($n = 6$, $\bar x = 5.667$, $S_{xx} = 45.33$, $MSE = 1.92/4 = 0.48$, $\hat y = 2.37 + 1.346x$), at $x = 7$: $\hat y = 11.79$, $\frac{1}{6} + \frac{(1.333)^2}{45.33} = 0.206$, and $t_{0.025}(4) = 2.776$. The 95% confidence interval for the mean is $11.79 \pm 2.776\sqrt{0.48 \cdot 0.206} = 11.79 \pm 0.87$, and the prediction interval is $11.79 \pm 2.776\sqrt{0.48 \cdot 1.206} = 11.79 \pm 2.11$.

## 4.3 Inference on the correlation

When $(X, Y)$ is bivariate normal with correlation $\rho$, the conditional mean is linear with slope $\beta_1 = \rho\sigma_Y/\sigma_X$, so $\rho = 0 \iff \beta_1 = 0$.

**Theorem 4.6**

(a) The $t$ statistic for $\beta_1 = 0$ equals $t_0 = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$, which is $t(n-2)$ under $\rho = 0$.

(b) (Fisher) $Z = \frac{1}{2}\log\frac{1+r}{1-r}$ is approximately $N\big(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\big)$ for moderate $n$ (say $n \geq 25$).

**Proof.** (a) $SSE = SST(1-r^2)$ and $\hat\beta_1^2 S_{xx} = SSR = r^2 SST$, so

$$t_0^2 = \frac{\hat\beta_1^2 S_{xx}}{SSE/(n-2)} = \frac{(n-2)r^2}{1-r^2}.$$

The null distribution is conditionally $t(n-2)$ given the $x_i$, hence unconditionally as well.

(b) is the variance-stabilizing transformation of the asymptotic law $\sqrt{n}(r-\rho) \to N(0, (1-\rho^2)^2)$; the correction $n-3$ improves the approximation. ■

Part (b) gives tests of $H_0 : \rho = \rho_0 \neq 0$ and confidence intervals: $\tanh\big(Z \pm z_{\alpha/2}/\sqrt{n-3}\big)$ covers $\rho$ with probability about $1-\alpha$.

## 4.4 Residuals

Residuals are our window on the errors. In vector form $\mathbf{e} = (I-H)\mathbf{y}$ with $H$ the hat matrix (Chapter 6), and:

**Theorem 4.7**

$Ee_i = 0$ and

$$\operatorname{Var}(e_i) = \sigma^2\Big(1 - \frac{1}{n} - \frac{(x_i-\bar x)^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2\Big(\frac{1}{n} + \frac{(x_i-\bar x)(x_j-\bar x)}{S_{xx}}\Big) \ (i \neq j).$$

**Proof.** $e_i = y_i - \bar y - \hat\beta_1(x_i - \bar x)$. Using $\operatorname{Var}\bar y = \sigma^2/n$, $\operatorname{Var}\hat\beta_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(y_i, \bar y) = \sigma^2/n$, $\operatorname{Cov}(y_i, \hat\beta_1) = \sigma^2 k_i$ and $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$,

$$\operatorname{Var} e_i = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}} - \frac{2}{n} - 2\frac{(x_i-\bar x)^2}{S_{xx}}\Big),$$

which simplifies to the claim; the covariance is computed in the same way. ■

Thus residuals are neither independent nor equally variable; points far from $\bar x$ have smaller residual variance (Chapter 10). Since $\sum e_i = 0$ and $\sum x_i e_i = 0$, residual plots against $x_i$ or $\hat y_i$ have no linear trend by construction; what we look for is *non-random* structure:

- a curved band suggests a missing quadratic term or a transformation;
- a funnel suggests nonconstant variance (weighted least squares or a variance-stabilizing transformation of $y$);
- a trend in time order suggests correlated errors (Chapter 12).

# Chapter 5. Transformations, Comparisons and Quadratic Forms

## 5.1 Transformations and the Box–Cox family

When the scatter plot or the residual plot shows curvature or a funnel shape, a transformation of $y$ (or $x$) can restore linearity, constant variance and normality at once. Common choices are $\log y$, $\sqrt{y}$ and $1/y$. Box and Cox embedded them in one family:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log y, & \lambda = 0, \end{cases} \qquad y > 0,$$

which is continuous in $\lambda$ since $(y^\lambda - 1)/\lambda \to \log y$. $\lambda = 1$ is (essentially) no transformation, $\lambda = \frac{1}{2}$ the square root and $\lambda = -1$ the reciprocal. The model is

$$y_i^{(\lambda)} = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

with $\lambda$ an additional parameter.

**Choosing $\lambda$.** Comparing $SSE$ across $\lambda$ directly is meaningless, because the scale of $y^{(\lambda)}$ changes with $\lambda$. Divide by the Jacobian: with the geometric mean $\dot{y} = (\prod y_i)^{1/n}$, set

$$z_i^{(\lambda)} = \begin{cases} \dfrac{y_i^\lambda - 1}{\lambda \dot{y}^{\lambda-1}}, & \lambda \neq 0, \\ \dot{y} \log y_i, & \lambda = 0. \end{cases}$$

Then fit $z^{(\lambda)} = \beta_0 + \beta_1 x + \epsilon$ for a grid of $\lambda$ and choose the $\lambda$ with the smallest $SSE(\lambda)$.

**Proposition 5.1**

The profile log-likelihood of $\lambda$ in the Box–Cox model is $-\frac{n}{2}\log SSE(\lambda) + \text{const}$, where $SSE(\lambda)$ is the residual sum of squares of the regression of $z^{(\lambda)}$. Hence minimizing $SSE(\lambda)$ is maximum likelihood estimation of $\lambda$.

**Proof.** The density of $y_i$ is the normal density of $y_i^{(\lambda)}$ times the Jacobian $y_i^{\lambda-1}$. Maximizing over $\beta$ and $\sigma^2$ for fixed $\lambda$ gives

$$l_{\max}(\lambda) = -\frac{n}{2}\log\frac{SSE_y(\lambda)}{n} - \frac{n}{2} + (\lambda - 1)\sum \log y_i,$$

where $SSE_y(\lambda)$ is the residual sum of squares for $y^{(\lambda)}$. Since $z^{(\lambda)} = y^{(\lambda)}/\dot{y}^{\lambda-1}$, $SSE(\lambda) = SSE_y(\lambda)\dot{y}^{-2(\lambda-1)}$, and $(\lambda-1)\sum \log y_i = n(\lambda-1)\log \dot{y}$. Substituting, $l_{\max}(\lambda) = -\frac{n}{2}\log SSE(\lambda) + \text{const}$. ∎

An approximate 95% confidence set for $\lambda$ is $\{\lambda : 2\{l_{\max}(\hat{\lambda}) - l_{\max}(\lambda)\} \le \chi^2_{0.05}(1)\}$; in practice one picks a simple value $(0, \frac{1}{2}, 1)$ inside it.

## 5.2 Measurement error in the explanatory variable

Suppose the true model is $y_i = \beta_0 + \beta_1 x_i^* + \epsilon_i$, but we observe $x_i = x_i^* + \delta_i$, where $\delta_i$ has mean 0 and variance $\sigma_\delta^2$ and is independent of $x_i^*$ and $\epsilon_i$. Treat the $x_i^*$ as random with variance $\sigma_*^2$. Then $y_i = \beta_0 + \beta_1 x_i + (\epsilon_i - \beta_1\delta_i)$, and the new error is correlated with the regressor $x_i$.

**Proposition 5.2 (Attenuation)**

The least squares slope from regressing $y$ on the observed $x$ satisfies

$$\hat{\beta}_1 \xrightarrow{p} \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} = \beta_1 \frac{\sigma_*^2}{\sigma_*^2+\sigma_\delta^2} = \beta_1\Big(1-\frac{\sigma_\delta^2}{\mathrm{Var}\,x}\Big).$$

**Proof.** $\hat{\beta}_1 = \frac{\frac{1}{n}\sum(x_i-\bar{x})(y_i-\bar{y})}{\frac{1}{n}\sum(x_i-\bar{x})^2}$, a ratio of sample moments that converges to $\mathrm{Cov}(x,y)/\mathrm{Var}(x)$. Here $\mathrm{Cov}(x,y) = \mathrm{Cov}(x^*+\delta, \beta_1 x^* + \epsilon) = \beta_1\sigma_*^2$ and $\mathrm{Var}\,x = \sigma_*^2+\sigma_\delta^2$. ■

So the slope is biased towards zero (**attenuation** or regression dilution). The bias is small when $\sigma_\delta^2$ is small relative to the spread of the $x$'s. If the experimenter *sets* the observed value $x$ and the actual value fluctuates around it (Berkson's model, $x^* = x+\delta$), the error is independent of $x$ and least squares remains unbiased.

## 5.3 Comparing two regression lines

Two groups, $i = 1, 2$, give

$$y_{ij} = \beta_{0i} + \beta_{1i}x_{ij} + \epsilon_{ij}, \qquad \epsilon_{ij} \overset{\text{iid}}{\sim} N(0,\sigma^2), \qquad j = 1,\dots,n_i,$$

with a *common* error variance. This is the **full model**.

**Are the lines identical?** Test $H_0 : \beta_{01} = \beta_{02},\ \beta_{11} = \beta_{12}$. Under $H_0$ we have the **reduced model** $y_{ij} = \beta_0 + \beta_1 x_{ij} + \epsilon_{ij}$ for the pooled $n_1+n_2$ observations.

(1) Fit each group separately: $SSE(F) = SSE_1 + SSE_2$, with $df_F = n_1+n_2-4$.

(2) Fit one line to the pooled data: $SSE(R)$, with $df_R = n_1+n_2-2$.

(3) Compute

$$F_0 = \frac{\{SSE(R)-SSE(F)\}/(df_R-df_F)}{SSE(F)/df_F} = \frac{\{SSE(R)-SSE(F)\}/2}{SSE(F)/(n_1+n_2-4)}.$$

(4) Reject $H_0$ when $F_0 > F_\alpha(2, n_1+n_2-4)$.

That $F_0 \sim F(2, n_1+n_2-4)$ under $H_0$ follows from the general theory of Chapters 7 and 8: $SSE(F)/\sigma^2 \sim \chi^2(df_F)$, and $\{SSE(R)-SSE(F)\}/\sigma^2 \sim \chi^2(2)$ independently (a general linear hypothesis with two constraints).

**Are the lines parallel?** For $H_0 : \beta_{11} = \beta_{12}$, the separate slope estimates are independent, so

$$\mathrm{Var}(\hat{\beta}_{11}-\hat{\beta}_{12}) = \sigma^2\Big(\frac{1}{S_{xx,1}}+\frac{1}{S_{xx,2}}\Big), \qquad t_0 = \frac{\hat{\beta}_{11}-\hat{\beta}_{12}}{\sqrt{MSE(F)\big(\frac{1}{S_{xx,1}}+\frac{1}{S_{xx,2}}\big)}} \sim t(n_1+n_2-4)$$

under $H_0$, with $MSE(F) = SSE(F)/(n_1+n_2-4)$. Reject when $|t_0| > t_{\alpha/2}(n_1+n_2-4)$. The same comparisons can be done in a single regression with indicator variables (Chapter 13).

## 5.4 Multivariate normal distribution

The distribution theory of regression rests on the multivariate normal distribution and quadratic forms.

**Definition (Multivariate normal)**

$\mathbf{X} \sim N_n(\mu, \Sigma)$, with $\Sigma$ symmetric nonnegative definite (possibly singular), if any of the following equivalent conditions holds:

(1) $\mathbf{X} \overset{d}{=} A\mathbf{Z} + \mu$ for some $n \times k$ matrix $A$ with $AA' = \Sigma$ and $\mathbf{Z} \sim N_k(\mathbf{0}, I)$;

(2) $\mathbf{X} \overset{d}{=} \Sigma^{1/2}\mathbf{Z} + \mu$ with $\mathbf{Z} \sim N_n(\mathbf{0}, I)$;

(3) $Ee^{\mathbf{t}'\mathbf{X}} = \exp(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$ for all $\mathbf{t} \in \mathbb{R}^n$;

(4) $\mathbf{a}'\mathbf{X} \sim N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$ for every $\mathbf{a}$.

The equivalences follow from $Ee^{\mathbf{s}'\mathbf{Z}} = e^{\mathbf{s}'\mathbf{s}/2}$: for instance, (1) gives $Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu}e^{\frac{1}{2}\mathbf{t}'AA'\mathbf{t}}$, which is (3). Linear maps preserve normality: $B\mathbf{X} + \mathbf{c} \sim N(B\mu + \mathbf{c}, B\Sigma B')$, and jointly normal vectors are independent iff they are uncorrelated.

## 5.5 Quadratic forms

Noncentral chi-square: if $X_i \sim N(\mu_i, 1)$ are independent, $\sum_{i=1}^r X_i^2 \sim \chi^2(r, \lambda)$ with noncentrality $\lambda = \frac{1}{2}\sum \mu_i^2$. (Some books use $\sum \mu_i^2$ instead of half of it; we follow the half convention of the lecture.)

**Theorem 5.3**

If $\mathbf{Z} \sim N(\mathbf{0}, I)$ and $A$ is symmetric with $A^2 = A$, then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$ with $r = \mathrm{tr}A$. Conversely, if $\mathbf{Z}'A\mathbf{Z}$ is chi-square, then $A^2 = A$.

**Proof.** Diagonalize $A = P\,\mathrm{diag}(\lambda_i)P'$ with $P$ orthogonal. $A^2 = A$ forces $\lambda_i \in \{0, 1\}$, and the number of ones is $\mathrm{tr}A = r$. With $\mathbf{W} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$, $\mathbf{Z}'A\mathbf{Z} = \sum_i \lambda_i W_i^2$, a sum of $r$ squared independent standard normals. For the converse, compare the mgfs $\prod_i (1 - 2\lambda_i t)^{-1/2}$ and $(1 - 2t)^{-r/2}$: equality of the polynomials $\prod(1 - 2\lambda_i t)$ and $(1 - 2t)^r$ forces each $\lambda_i \in \{0, 1\}$. ∎

**Theorem 5.4 (Moments of quadratic forms)**

Let $\mathbf{y} \sim N(\mu, V)$ and $A$ symmetric. Then

(a) $E(\mathbf{y}'A\mathbf{y}) = \mathrm{tr}(AV) + \mu'A\mu$ (this needs only the mean and covariance);

(b) $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = 2VA\mu$;

(c) $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = 2\mathrm{tr}\{(AV)^2\} + 4\mu'AVA\mu$.

**Proof.** (a) $\mathbf{y}'A\mathbf{y} = \mathrm{tr}(A\mathbf{y}\mathbf{y}')$ and $E\mathbf{y}\mathbf{y}' = V + \mu\mu'$. (b), (c) Write $\mathbf{y} = \mu + \mathbf{w}$ with $\mathbf{w} \sim N(\mathbf{0}, V)$: $\mathbf{y}'A\mathbf{y} = \mathbf{w}'A\mathbf{w} + 2\mu'A\mathbf{w} + \mu'A\mu$. Odd moments of $\mathbf{w}$ vanish, so $\mathrm{Cov}(\mathbf{w}, \mathbf{w}'A\mathbf{w}) = 0$ and $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = \mathrm{Cov}(\mathbf{w}, 2\mu'A\mathbf{w}) = 2VA\mu$. Likewise $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = \mathrm{Var}(\mathbf{w}'A\mathbf{w}) + 4\mu'AVA\mu$. Write $\mathbf{w} = V^{1/2}\mathbf{z}$ and diagonalize $V^{1/2}AV^{1/2}$ with eigenvalues $\lambda_i$: then $\mathrm{Var}(\mathbf{w}'A\mathbf{w}) = 2\sum \lambda_i^2 = 2\mathrm{tr}\{(AV)^2\}$. ∎

**Theorem 5.5**

Let $\mathbf{y} \sim N(\mu, V)$ with $V$ nonsingular and $A$ symmetric.

(a) $\mathbf{y}'A\mathbf{y} \sim \chi^2(r(A), \frac{1}{2}\mu'A\mu)$ if and only if $AV$ is idempotent.

(b) In particular, if $\mathbf{y} \sim N(\mu, \sigma^2 I)$, then $\mathbf{y}'A\mathbf{y}/\sigma^2 \sim \chi^2(r(A), \frac{1}{2\sigma^2}\mu'A\mu)$ iff $A$ is idempotent; and $\mathbf{y}'\mathbf{y}/\sigma^2 \sim \chi^2(n, \mu'\mu/(2\sigma^2))$.

**Proof of sufficiency in (a).** Let $\mathbf{z} = V^{-1/2}\mathbf{y} \sim N(V^{-1/2}\mu, I)$ and $B = V^{1/2}AV^{1/2}$. Then $\mathbf{y}'A\mathbf{y} = \mathbf{z}'B\mathbf{z}$, and $B^2 = V^{1/2}AVAV^{1/2} = V^{1/2}AV^{1/2} = B$ when $AVA = A$, which follows from $(AV)^2 = AV$. Diagonalize $B = P\,\mathrm{diag}(I_r, 0)P'$; with $\mathbf{w} = P'\mathbf{z} \sim N(P'V^{-1/2}\mu, I)$, $\mathbf{z}'B\mathbf{z} = \sum_{i\le r} w_i^2$, a noncentral chi-square with $r = \mathrm{tr}B = r(A)$ degrees of freedom and noncentrality $\frac{1}{2}\sum_{i\le r}(Ew_i)^2 = \frac{1}{2}\mu'V^{-1/2}BV^{-1/2}\mu = \frac{1}{2}\mu'A\mu$. Necessity follows by comparing mgfs as in Theorem 5.3. ■

**Theorem 5.6 (Independence)**

Let $\mathbf{y} \sim N(\mu, V)$.

(a) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent if $BVA = 0$.

(b) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ (with $A, B$ symmetric) are independent if $AVB = 0$.

(The conditions are also necessary.)

**Proof.** We give the proof when $A$ and $B$ are nonnegative definite (for instance idempotent), the case used in regression.

(a) Write $A = LL'$ with $L$ of full column rank. $L'\mathbf{y}$ and $B\mathbf{y}$ are jointly normal with $\mathrm{Cov}(B\mathbf{y}, L'\mathbf{y}) = BVL$. From $BVA = BVLL' = 0$, multiplying by $L(L'L)^{-1}$ on the right gives $BVL = 0$, so $L'\mathbf{y}$ and $B\mathbf{y}$ are independent, and $\mathbf{y}'A\mathbf{y} = \|L'\mathbf{y}\|^2$ is a function of $L'\mathbf{y}$.

(b) The same argument applied to both forms: $\mathrm{Cov}(L_A'\mathbf{y}, L_B'\mathbf{y}) = L_A'VL_B = 0$ when $AVB = 0$. ■

**Theorem 5.7 (Cochran)**

Let $\mathbf{y} \sim N(\mathbf{0}, I_n)$ and let $A_1, \dots, A_p$ be symmetric with $\sum A_j = I_n$ and $r(A_j) = k_j$. Then the forms $\mathbf{y}'A_j\mathbf{y}$ are independent with $\mathbf{y}'A_j\mathbf{y} \sim \chi^2(k_j)$ if and only if $\sum k_j = n$.

**Proof of sufficiency.** Since $\sum A_j = I$ and $\sum r(A_j) = n$, the column spaces of the $A_j$ together span $\mathbb{R}^n$ with no overlap: $\mathbb{R}^n = \bigoplus_j \mathrm{col}(A_j)$. For $\mathbf{v} \in \mathrm{col}(A_k)$, $\mathbf{v} = \sum_j A_j\mathbf{v}$ with $A_j\mathbf{v} \in \mathrm{col}(A_j)$; uniqueness of the decomposition gives $A_k\mathbf{v} = \mathbf{v}$ and $A_j\mathbf{v} = \mathbf{0}$ for $j \ne k$. Hence $A_k^2 = A_k$ and $A_jA_k = 0$ $(j \ne k)$. Theorems 5.3 and 5.6 finish the proof. ■

## 5.6 Expected mean squares in simple regression

In the centred model $y_i = \beta_0^* + \beta_1(x_i - \bar{x}) + \epsilon_i$:

**Proposition 5.8**

$$E(SST) = \beta_1^2 S_{xx} + (n-1)\sigma^2, \qquad E(SSR) = \sigma^2 + \beta_1^2 S_{xx}, \qquad E(SSE) = (n-2)\sigma^2.$$

**Proof.** $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\epsilon_i - \bar{\epsilon})$, so $E(SST) = \beta_1^2 S_{xx} + E\sum(\epsilon_i - \bar{\epsilon})^2 = \beta_1^2 S_{xx} + (n-1)\sigma^2$ (the cross term has mean 0). $E(SSR) = S_{xx}E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Subtract. ■

In matrix form, with $J_n = \mathbf{1}\mathbf{1}'$, $SST = \mathbf{y}'(I - J_n/n)\mathbf{y}$; $I - J_n/n$ is idempotent of rank $n-1$, and Theorem 5.4(a) reproduces $E(SST)$.

# Chapter 6. Multiple Linear Regression

## 6.1 The model in matrix form

With $p$ explanatory variables,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n,$$

or, stacking the observations,

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

where $\mathbf{y}$ is $n \times 1$, $X$ is the $n \times (p+1)$ **design matrix** whose $i$-th row is $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$, and $\beta = (\beta_0, \ldots, \beta_p)'$. We assume $n > p+1$ and $\operatorname{rank} X = p+1$. Simple regression is the case $p = 1$, with rows $(1, x_i)$.

## 6.2 Least squares

**Theorem 6.1**

The least squares estimator minimizing $S(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$ is the solution of the normal equations $X'X\hat\beta = X'\mathbf{y}$:

$$\hat\beta = (X'X)^{-1}X'\mathbf{y}.$$

**Proof.** $S(\beta) = \mathbf{y}'\mathbf{y} - 2\beta'X'\mathbf{y} + \beta'X'X\beta$ has gradient $-2X'\mathbf{y} + 2X'X\beta$ and Hessian $2X'X$, which is positive definite because $X$ has full column rank ($\mathbf{a}'X'X\mathbf{a} = \|X\mathbf{a}\|^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$). So the unique stationary point is the global minimum. Alternatively, for any $\beta$,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat\beta\|^2 + \|X(\hat\beta - \beta)\|^2,$$

because the cross term $2(\hat\beta - \beta)'X'(\mathbf{y} - X\hat\beta)$ vanishes by the normal equations. ∎

For $p = 1$,

$$X'X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix},$$

which reproduces the formulas of Chapter 3.

**Example 6.1**

For $n = 6$ observations of $(x_1, x_2, y)$:

| | | | | | | |
|---|---|---|---|---|---|---|
| $x_1$ | 1 | 2 | 3 | 4 | 5 | 6 |
| $x_2$ | 2 | 1 | 4 | 3 | 6 | 5 |
| $y$ | 3 | 6 | 7 | 10 | 12 | 13 |

Centring gives $\bar x_1 = \bar x_2 = 3.5$, $\bar y = 8.5$, and the centred cross products $S_{11} = S_{22} = 17.5$, $S_{12} = 14.5$, $S_{1y} = 35.5$, $S_{2y} = 28.5$. The slopes solve

$$\begin{pmatrix} 17.5 & 14.5 \\ 14.5 & 17.5 \end{pmatrix} \begin{pmatrix} \hat\beta_1 \\ \hat\beta_2 \end{pmatrix} = \begin{pmatrix} 35.5 \\ 28.5 \end{pmatrix},$$

so, with determinant $17.5^2 - 14.5^2 = 96$,

$$\hat\beta_1 = \frac{17.5 \cdot 35.5 - 14.5 \cdot 28.5}{96} = 2.167, \qquad \hat\beta_2 = \frac{17.5 \cdot 28.5 - 14.5 \cdot 35.5}{96} = -0.167,$$

and $\hat\beta_0 = 8.5 - 3.5(2.167 - 0.167) = 1.5$. Note that $y$ is strongly positively correlated with $x_2$ on its own ($S_{2y} > 0$), yet the coefficient of $x_2$ is slightly negative once $x_1$ is in the model: $x_2$ carries little information beyond $x_1$ (Section 6.6).

The **fitted values** are $\hat{\mathbf{y}} = X\hat\beta = H\mathbf{y}$ with the **hat matrix** $H = X(X'X)^{-1}X'$, and the residuals are $\mathbf{e} = (I - H)\mathbf{y}$. $H$ is symmetric and idempotent with $HX = X$ and $\operatorname{tr} H = p + 1$: it is the orthogonal projection onto the column space of $X$. Hence $X'\mathbf{e} = \mathbf{0}$: the residuals sum to zero (the intercept column) and are orthogonal to every regressor and to $\hat{\mathbf{y}}$.

## 6.3 Analysis of variance

With $J_n = \mathbf{1}\mathbf{1}'$, $n\bar y^2 = \mathbf{y}'(J_n/n)\mathbf{y}$ is the **correction term**. The sums of squares are quadratic forms:

$$SST = \mathbf{y}'\Big(I - \frac{J_n}{n}\Big)\mathbf{y},$$

$$SSE = \mathbf{y}'(I - H)\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat\beta' X'\mathbf{y}, \qquad SSR = \mathbf{y}'\Big(H - \frac{J_n}{n}\Big)\mathbf{y} = \hat\beta' X'\mathbf{y} - n\bar y^2,$$

with $SST = SSR + SSE$. The degrees of freedom are the ranks of the matrices: $n - 1$, $p$ and $n - p - 1$. ($SST$ has $n - 1$ degrees of freedom because the $n$ deviations $y_i - \bar y$ satisfy one linear constraint.)

| source | SS | df | MS | $F_0$ |
|---|---|---|---|---|
| regression | $SSR$ | $p$ | $MSR = SSR/p$ | $MSR/MSE$ |
| residual | $SSE$ | $n - p - 1$ | $MSE = SSE/(n - p - 1)$ | |
| total | $SST$ | $n - 1$ | | |

**Theorem 6.2 (Expected mean squares)**

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + \frac{1}{p}\beta' X'\Big(I - \frac{J_n}{n}\Big)X\beta.$$

The second term is $\geq 0$, with equality iff $\beta_1 = \cdots = \beta_p = 0$.

**Proof.** By Theorem 5.4(a) with $E\mathbf{y} = X\beta$ and $\operatorname{Var}\mathbf{y} = \sigma^2 I$: for $B = I - H$, $\operatorname{tr} B = n - (p+1)$ and $X'BX = 0$, so $E(SSE) = (n - p - 1)\sigma^2$. For $A = H - J_n/n$, $\operatorname{tr} A = (p + 1) - 1 = p$ and $X'AX = X'X - X'J_nX/n = X'(I - J_n/n)X$, since $HX = X$. Finally $\beta' X'(I - J_n/n)X\beta = \|(I - J_n/n)X\beta\|^2$ is the squared length of the centred mean vector, which is zero iff $X\beta$ is constant, i.e. iff all slopes vanish (the centred columns of $X$ are linearly independent). ■

For $p = 1$, $\beta' X'(I - J_n/n)X\beta = \beta_1^2 S_{xx}$, recovering Chapter 5. The **coefficient of determination** is $R^2 = SSR/SST = 1 - SSE/SST$, and $\hat\sigma^2 = MSE$ is the **residual mean square**.

## 6.4 Properties of the estimators

**Theorem 6.3**

(a) $E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}$; writing $(X'X)^{-1} = (c_{ij})$, $\operatorname{Var}\hat\beta_j = c_{jj}\sigma^2$ and $\operatorname{Cov}(\hat\beta_i, \hat\beta_j) = c_{ij}\sigma^2$.

(b) At a point $\mathbf{x} = (1, x_1, \ldots, x_p)'$, $\hat y = \mathbf{x}'\hat\beta$ estimates $E(y \mid \mathbf{x})$ with $\operatorname{Var}\hat y = \mathbf{x}'(X'X)^{-1}\mathbf{x}\,\sigma^2$.

(c) A new observation $y_s$ at $\mathbf{x}$ has $\mathrm{Var}(y_s - \hat{y}) = \{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}\sigma^2$.

**Proof.** (a) $E\hat{\beta} = (X'X)^{-1}X'X\beta = \beta$ and $\mathrm{Var}\,\hat{\beta} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $\mathrm{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}'\,\mathrm{Var}(\hat{\beta})\mathbf{x}$.

(c) $y_s$ is independent of $\hat{y}$. ■

For $p = 1$ these reduce to $\mathrm{Var}\,\hat{\beta}_0 = \sigma^2(1/n + \bar{x}^2/S_{xx})$, $\mathrm{Var}\,\hat{\beta}_1 = \sigma^2/S_{xx}$, $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ and $\mathrm{Var}\,\hat{y} = \sigma^2\{1/n + (x - \bar{x})^2/S_{xx}\}$, so $\hat{\beta}_0$ and $\hat{\beta}_1$ are uncorrelated iff $\bar{x} = 0$. The precision of $\hat{\beta}$ depends on the design through $(X'X)^{-1}$; choosing the design well (spreading the $x$'s, avoiding near-collinear columns) reduces the variances.

## 6.5 Distribution theory

**Theorem 6.4**

Under $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$:

(a) $SSE/\sigma^2 \sim \chi^2(n - p - 1)$;

(b) $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ with $\lambda = \beta'X'(I - J_n/n)X\beta/(2\sigma^2)$;

(c) $SST/\sigma^2 \sim \chi^2(n - 1, \lambda)$ with the same $\lambda$;

(d) $SSR$ and $SSE$ are independent, and $\hat{\beta}$ is independent of $SSE$;

(e) $F_0 = MSR/MSE \sim F(p, n - p - 1, \lambda)$; under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $F_0 \sim F(p, n - p - 1)$.

**Proof.** All three matrices $I - H$, $H - J_n/n$ and $I - J_n/n$ are symmetric idempotent: for $H - J_n/n$ use $HJ_n = J_nH = J_n$ (since $H\mathbf{1} = \mathbf{1}$, the intercept column) and $J_n^2 = nJ_n$:

$$\left(H - \frac{J_n}{n}\right)^2 = H - \frac{J_n}{n} - \frac{J_n}{n} + \frac{J_n}{n} = H - \frac{J_n}{n}.$$

Their ranks (traces) are $n - p - 1$, $p$ and $n - 1$. Theorem 5.5(b) gives (a)–(c), the noncentralities being $\frac{1}{2\sigma^2}(X\beta)'A(X\beta)$; for $I - H$ it is 0 because $(I - H)X = 0$.

(d) $(H - J_n/n)(I - H) = 0$, so Theorem 5.6(b) gives independence of $SSR$ and $SSE$. Also $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ and $(X'X)^{-1}X'(I - H) = 0$, so Theorem 5.6(a) applies.

(e) follows from (a), (b), (d). ■

Consequently, with $c_{jj}$ the diagonal of $(X'X)^{-1}$,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{c_{jj}MSE}} \sim t(n - p - 1).$$

**Example 6.2**

In Example 6.1, $SST = \sum(y_i - 8.5)^2 = 73.5$ and $SSR = \hat{\beta}_1 S_{1y} + \hat{\beta}_2 S_{2y} = 2.167 \cdot 35.5 - 0.167 \cdot 28.5 = 72.17$, so $SSE = 1.33$ with 3 degrees of freedom and $R^2 = 0.982$. Then $F_0 = (72.17/2)/(1.33/3) = 81.2 > F_{0.05}(2, 3) = 9.55$: the regression is highly significant.

## 6.6 Partial regression: the Frisch–Waugh–Lovell theorem

Partition $X = (X_1, X_2)$, with $X_1$ $(n \times (q+1))$ containing the intercept and $X_2$ $(n \times (p-q))$ the remaining columns, and write $\mathbf{y} = X_1\beta_1 + X_2\beta_2 + \epsilon$. Let $H_1 = X_1(X_1'X_1)^{-1}X_1'$.

**Theorem 6.5 (Frisch–Waugh–Lovell)**

Let $\mathbf{e} = (I - H_1)\mathbf{y}$ be the residuals of regressing $\mathbf{y}$ on $X_1$, and $X_{2\cdot1} = (I - H_1)X_2$ the residuals of regressing each column of $X_2$ on $X_1$. The least squares coefficient from regressing $\mathbf{e}$ on $X_{2\cdot1}$,

$$\tilde{\beta}_2 = (X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e},$$

equals the $\hat{\beta}_2$ part of the full least squares estimator. Moreover the residuals of the two regressions coincide.

**Proof.** The normal equations of the full model are

$$\begin{pmatrix} X_2'X_2 & X_2'X_1 \\ X_1'X_2 & X_1'X_1 \end{pmatrix} \begin{pmatrix} \hat{\beta}_2 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} X_2'\mathbf{y} \\ X_1'\mathbf{y} \end{pmatrix}.$$

Multiply on the left by $\begin{pmatrix} I & -X_2'X_1(X_1'X_1)^{-1} \\ 0 & I \end{pmatrix}$. The first block row becomes

$$X_2'(I - H_1)X_2\hat{\beta}_2 + \{X_2'X_1 - X_2'H_1X_1\}\hat{\beta}_1 = X_2'(I - H_1)\mathbf{y},$$

and the bracket is 0 since $H_1X_1 = X_1$. Hence $\hat{\beta}_2 = \{X_2'(I - H_1)X_2\}^{-1}X_2'(I - H_1)\mathbf{y}$. Because $I - H_1$ is symmetric idempotent, this is $(X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e} = \tilde{\beta}_2$. For the residuals: the full fit is $X_1\hat{\beta}_1 + X_2\hat{\beta}_2$; applying $I - H_1$ to $\mathbf{y} - X_1\hat{\beta}_1 - X_2\hat{\beta}_2$ (which is orthogonal to $X_1$) gives $\mathbf{e} - X_{2\cdot1}\hat{\beta}_2$, the residual of the second regression. ■

**Interpretation**

A multiple regression coefficient $\hat{\beta}_j$ measures the effect of $x_j$ on $y$ *after both have been adjusted for the other regressors*: it is the slope of the residuals of $y$ against the residuals of $x_j$. It is generally different from the simple regression slope of $y$ on $x_j$, and the two agree when $x_j$ is orthogonal to the other (centred) regressors.

# Chapter 7. Inference in Multiple Regression I

Throughout, $\mathbf{y} = X\beta + \epsilon$ with $X$ of size $n \times (p+1)$ and full column rank. We distinguish two sets of assumptions:

- **(A)** $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$, needed for exact $t$ and $F$ inference;
- **(B)** $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I_n$ only, enough for the optimality of least squares.

## 7.1 The Gauss–Markov theorem

**Theorem 7.1 (Gauss–Markov)**

Under (B), among all linear unbiased estimators $\tilde\beta = A\mathbf{y}$ of $\beta$, the least squares estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ has the smallest variance: $\operatorname{Var}\tilde\beta - \operatorname{Var}\hat\beta$ is nonnegative definite. Equivalently, for every $\mathbf{c}$, $\mathbf{c}'\hat\beta$ has the smallest variance among linear unbiased estimators of $\mathbf{c}'\beta$.

**Proof.** $E(A\mathbf{y}) = AX\beta = \beta$ for all $\beta$ forces $AX = I$. Write $A = (X'X)^{-1}X' + D$; then $DX = AX - I = 0$. Hence

$$\operatorname{Var}(A\mathbf{y}) = \sigma^2 AA' = \sigma^2\{(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD'\} = \sigma^2(X'X)^{-1} + \sigma^2 DD',$$

and $DD'$ is nonnegative definite, with $\mathbf{c}'DD'\mathbf{c} = \|D'\mathbf{c}\|^2$. Equality for all $\mathbf{c}$ holds iff $D = 0$. ■

So $\hat\beta$ is the **best linear unbiased estimator** (BLUE). The theorem needs neither normality nor independence, only uncorrelated errors with a common variance. If $\operatorname{Var}\epsilon = \sigma^2 V$ with $V \neq I$, the BLUE is the generalized least squares estimator (Chapter 12).

## 7.2 Confidence intervals

Under (A), with $(X'X)^{-1} = (c_{ij})$ and $MSE = SSE/(n-p-1)$ independent of $\hat\beta$ (Theorem 6.4):

**Theorem 7.2**

$100(1-\alpha)\%$ confidence intervals are:

(a) for a coefficient: $\hat\beta_j \pm t_{\alpha/2}(n-p-1)\sqrt{c_{jj}MSE}$;

(b) for a linear combination $\mathbf{q}'\beta$: $\mathbf{q}'\hat\beta \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{q}'(X'X)^{-1}\mathbf{q}\,MSE}$;

(c) for the mean response at $\mathbf{x}$: $\mathbf{x}'\hat\beta \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{x}'(X'X)^{-1}\mathbf{x}\,MSE}$;

(d) a prediction interval for a new observation at $\mathbf{x}$: $\mathbf{x}'\hat\beta \pm t_{\alpha/2}(n-p-1)\sqrt{\{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}MSE}$.

With $\sigma^2$ known, replace $MSE$ by $\sigma^2$ and $t_{\alpha/2}$ by $z_{\alpha/2}$.

**Proof.** (b) contains (a) and (c) as special cases. $\mathbf{q}'\hat\beta \sim N(\mathbf{q}'\beta, \sigma^2\mathbf{q}'(X'X)^{-1}\mathbf{q})$ is independent of $(n-p-1)MSE/\sigma^2 \sim \chi^2(n-p-1)$, so the studentized ratio is $t(n-p-1)$.

(d) adds the variance of the new error. ■

## 7.3 The general linear hypothesis

Many hypotheses are linear restrictions on $\beta$:

$$H_0 : C\beta = \mathbf{m} \quad \text{vs} \quad H_1 : C\beta \neq \mathbf{m},$$

with $C$ a known $k \times (p+1)$ matrix of rank $k$ and $\mathbf{m} \in \mathbb{R}^k$. Examples, for $E(y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_4 x_4$:

- $\beta_1 = \beta_2$ and $\beta_3 = 2\beta_4$: $C = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$, $\mathbf{m} = \mathbf{0}$;
- $\beta_1 = \beta_2 = \beta_3 = \beta_4$: three rows $(0, 1, -1, 0, 0)$, $(0, 0, 1, -1, 0)$, $(0, 0, 0, 1, -1)$;
- $\beta_j = 0$: $C = \mathbf{e}_j'$, the $j$-th unit row.

There are two equivalent routes: minimize $SSE$ under the restriction, or reparametrize to a reduced model.

### 7.3.1 Method I: constrained least squares

**Theorem 7.3 (Restricted least squares)**

The minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C\beta = \mathbf{m}$ is

$$\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}),$$

and its residual sum of squares is

$$SSE(\tilde{\beta}) = SSE + Q, \qquad Q = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}).$$

**Proof.** With a Lagrange multiplier vector $\lambda \in \mathbb{R}^k$, minimize $L = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta) + 2\lambda'(C\beta - \mathbf{m})$. Setting derivatives to zero:

$$X'X\tilde{\beta} + C'\lambda = X'\mathbf{y}, \qquad C\tilde{\beta} = \mathbf{m}.$$

The first gives $\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\lambda$, and substituting into the second, $C\hat{\beta} - C(X'X)^{-1}C'\lambda = \mathbf{m}$, so $\lambda = \{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m})$. ($C(X'X)^{-1}C'$ is invertible since $C$ has full row rank.) That this stationary point is the constrained minimum follows from the decomposition below. For any $\beta$,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta),$$

since $X'(\mathbf{y} - X\hat{\beta}) = \mathbf{0}$. With $\beta = \tilde{\beta}$ and $\hat{\beta} - \tilde{\beta} = (X'X)^{-1}C'\lambda$,

$$(\hat{\beta} - \tilde{\beta})'X'X(\hat{\beta} - \tilde{\beta}) = \lambda'C(X'X)^{-1}C'\lambda = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}) = Q.$$

For any feasible $\beta$, $(\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta) \geq Q$ (it is the squared distance, in the $X'X$ metric, from $\hat{\beta}$ to the affine set $\{C\beta = \mathbf{m}\}$, attained at $\tilde{\beta}$), so $\tilde{\beta}$ is optimal. ■

**Theorem 7.4 ($F$ test of the general linear hypothesis)**

Under (A),

$$\frac{Q}{\sigma^2} \sim \chi^2(k, \lambda), \qquad \lambda = \frac{(C\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\beta - \mathbf{m})}{2\sigma^2},$$

$Q$ is independent of $SSE$, and

$$F_0 = \frac{Q/k}{SSE/(n-p-1)} = \frac{Q}{k \cdot MSE} \sim F(k, n-p-1, \lambda).$$

Under $H_0$, $\lambda = 0$, and the test rejects when $F_0 > F_\alpha(k, n-p-1)$.

**Proof.** $C\hat{\beta} - \mathbf{m} \sim N(C\beta - \mathbf{m}, \sigma^2 C(X'X)^{-1}C')$, and $Q$ is the corresponding quadratic form with matrix $W = \{C(X'X)^{-1}C'\}^{-1}$. Since $W \cdot \sigma^2 C(X'X)^{-1}C'/\sigma^2 = I_k$ is idempotent, Theorem 5.5 gives the noncentral chi-square with $k$ degrees of freedom. $Q$ is a function of $\hat{\beta}$, which is independent of $SSE$ (Theorem 6.4). ■

### 7.3.2 Method II: full and reduced models

Substitute the restriction into the model to get a **reduced model** with $p+1-k$ free parameters; fit it and compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/k}{SSE(F)/(n-p-1)},$$

where $SSE(F) = SSE$ is from the full model, with $df_F = n-p-1$ and $df_R = n-p-1+k$. If the restriction involves a nonzero $\mathbf{m}$, move the known part to the left-hand side (an offset). Since $SSE(R)$ is exactly the constrained minimum $SSE(\tilde{\beta})$, Theorem 7.3 gives $SSE(R) - SSE(F) = Q$: *the two methods give the same statistic.*

For a model without intercept, $\mathbf{y} = X\beta + \epsilon$ with $X$ of size $n \times p$, the same holds with $n-p$ in place of $n-p-1$.

**Example 7.1**

A no-intercept model $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to

| | | | | | |
|---|---|---|---|---|---|
| $x_1$ | 1 | 2 | 0 | 1 | 3 |
| $x_2$ | 0 | 1 | 1 | 2 | 1 |
| $y$ | 3 | 7 | 2 | 5 | 10 |

Test $H_0 : \beta_1 = 2\beta_2$. Here

$$X'X = \begin{pmatrix} 15 & 7 \\ 7 & 7 \end{pmatrix}, \qquad X'\mathbf{y} = \begin{pmatrix} 52 \\ 29 \end{pmatrix}, \qquad \hat{\beta} = \frac{1}{56}\begin{pmatrix} 7 & -7 \\ -7 & 15 \end{pmatrix}\begin{pmatrix} 52 \\ 29 \end{pmatrix} = \begin{pmatrix} 2.875 \\ 1.268 \end{pmatrix},$$

$\mathbf{y}'\mathbf{y} = 187$, and $SSE = 187 - \hat{\beta}'X'\mathbf{y} = 0.732$ with 3 degrees of freedom, so $MSE = 0.244$. *Method I*: $C = (1, -2)$, $C\hat{\beta} = 0.339$, $C(X'X)^{-1}C' = \frac{7+28+60}{56} = 1.696$, so $Q = 0.339^2/1.696 = 0.068$ and $F_0 = 0.068/0.244 = 0.28$. *Method II*: under $H_0$, $y = \beta_2(2x_1 + x_2) + \epsilon$; with $z = 2x_1 + x_2 = (2, 5, 1, 4, 7)$, $\sum z^2 = 95$ and $\sum zy = 133$, so $SSE(R) = 187 - 133^2/95 = 0.800$ and $\{SSE(R) - SSE\}/MSE = 0.068/0.244 = 0.28$. Both give $F_0 = 0.28 < F_{0.05}(1, 3) = 10.13$: $H_0$ is not rejected.

## 7.4 Properties of residuals

**Theorem 7.5**

For the fitted model with intercept:

(a) $\sum_i e_i = 0$, $\sum_i x_{ij} e_i = 0$ for each $j$, and $\sum_i \hat{y}_i e_i = 0$;

(b) under (A), $\mathbf{e} = (I - H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I - H))$; in particular $\operatorname{Var} e_i = \sigma^2(1 - h_{ii})$ and $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

**Proof.** (a) is $X'\mathbf{e} = \mathbf{0}$ together with $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}'X'\mathbf{e}$.

(b) $E\mathbf{e} = (I - H)X\beta = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I - H)(I - H)' = \sigma^2(I - H)$. ■

So residuals are correlated, with correlation $-h_{ij}/\sqrt{(1 - h_{ii})(1 - h_{jj})}$, and have unequal variances depending on the design. Plots of residuals against $\hat{y}_i$, against each $x_{ij}$, and against time order are the basic checks:

- a horizontal band: no problem;
- a funnel: nonconstant variance; use weighted least squares or transform $y$;
- a linear trend against $x_j$: a computational error, since (a) forces zero correlation;
- a curve: add a quadratic term (or other function) of $x_j$, or transform.

# Chapter 8. Inference in Multiple Regression II

## 8.1 Extra sums of squares

When regressors are added to a model, $SSR$ increases (and $SSE$ decreases). The increase is the **extra sum of squares** due to the added regressors, and it measures their contribution *given* those already present.

For the full model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$, write

$$SS(\hat\beta_0, \ldots, \hat\beta_p) = \hat\beta' X'\mathbf{y} \quad (df = p+1), \qquad SS(\hat\beta_0) = n\bar y^2 \quad (df = 1),$$

$$SS(\hat\beta_1, \ldots, \hat\beta_p \mid \hat\beta_0) = SS(\hat\beta_0, \ldots, \hat\beta_p) - SS(\hat\beta_0) = SSR \quad (df = p).$$

If a submodel uses only $x_1, \ldots, x_q$ ($q < p$) with fit $\tilde\beta$, the extra sum of squares of $x_{q+1}, \ldots, x_p$ given $x_1, \ldots, x_q$ is

$$SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q) = SS(\hat\beta) - SS(\tilde\beta) = SSR(\text{full}) - SSR(\text{sub}) = SSE(\text{sub}) - SSE(\text{full}),$$

with $p - q$ degrees of freedom. In matrix form, partition $\beta = (\beta_1', \beta_2')'$ with $\beta_1 = (\beta_0, \ldots, \beta_q)'$, $X = (X_1, X_2)$, and let $H_1 = X_1(X_1'X_1)^{-1}X_1'$, $H = X(X'X)^{-1}X'$:

$$SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y}.$$

**Theorem 8.1**

(a) $H - H_1$ is symmetric idempotent of rank $p - q$, and $(H - H_1)(I - H) = 0$.

(b) Under normal errors, $SS(\beta_2 \mid \beta_1)/\sigma^2 \sim \chi^2(p-q, \lambda)$ with $\lambda = \beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$, independent of $SSE$.

(c) Hence, under $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$,

$$F_0 = \frac{SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q)/(p-q)}{MSE} \sim F(p-q, n-p-1).$$

**Proof.** (a) $HX = X$ gives $HX_1 = X_1$, hence $HH_1 = H_1$ and (transposing) $H_1H = H_1$. Then $(H-H_1)^2 = H - H_1 - H_1 + H_1 = H - H_1$, and $\operatorname{tr}(H - H_1) = (p+1) - (q+1)$. Also $(H - H_1)(I - H) = H - H - H_1 + H_1 = 0$.

(b) By Theorem 5.5(b) the noncentrality is $\frac{1}{2\sigma^2}(X\beta)'(H - H_1)X\beta$. Since $(H - H_1)X_1 = 0$ and $(H - H_1)X_2 = (I - H_1)X_2$, this equals $\beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$. Independence follows from (a) and Theorem 5.6.

(c) Under $H_0$, $\lambda = 0$. ■

This is the same test as the general linear hypothesis with $C = (0, I_{p-q})$ (Chapter 7), since $SSE(\text{sub}) - SSE(\text{full}) = Q$.

When the intercept is included in the regression sum of squares, one uses the uncorrected table: $SSR = \hat\beta' X'\mathbf{y}$ with $p + 1$ degrees of freedom and $SST = \mathbf{y}'\mathbf{y}$ with $n$; then $MSR/MSE \sim F(p + 1, n - p - 1)$ tests $\beta = \mathbf{0}$.

## 8.2 Partial and sequential $F$ tests

**Partial $F$ test.** The contribution of a single regressor $x_j$ given all others is $SS(\hat\beta_j \mid \text{all others})$, with one degree of freedom. The test of $H_0 : \beta_j = 0$,

$$F_0 = \frac{SS(\hat\beta_j \mid \hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p)}{MSE} \sim F(1, n-p-1),$$

is the **partial $F$ test**. It is equivalent to the $t$ test, since

$$SS(\hat\beta_j \mid \text{others}) = \frac{\hat\beta_j^2}{c_{jj}}, \qquad F_0 = \Big(\frac{\hat\beta_j}{\sqrt{c_{jj}MSE}}\Big)^2 = t_0^2.$$

(This follows from Theorem 7.3 with $C = \mathbf{e}_j'$: $Q = \hat\beta_j^2/c_{jj}$.) A non-significant partial $F$ means that $x_j$ adds little *once the other variables are in the model*; it does not mean that $x_j$ is unrelated to $y$. With correlated regressors, every partial $F$ can be small while the overall $F$ is large.

**Sequential $F$ tests.** Enter the regressors one at a time in a chosen order, and at each stage test the newly added variable given those already in:

$$SS(\hat\beta_1 \mid \hat\beta_0), \quad SS(\hat\beta_2 \mid \hat\beta_0, \hat\beta_1), \quad \ldots, \quad SS(\hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_{p-1}).$$

These sum to $SSR$, and each is compared with the $MSE$ of the model at that stage (or of the full model). The decomposition depends on the order unless the centred regressors are mutually orthogonal. The tests are valid only if the $MSE$ used is unbiased for $\sigma^2$, which requires that the omitted variables have no effect; otherwise the denominator is inflated.

## 8.3 Joint confidence regions and simultaneous intervals

**Theorem 8.2 (Joint confidence region)**

Under normal errors,

$$\frac{(\hat\beta-\beta)'X'X(\hat\beta-\beta)}{(p+1)MSE} \sim F(p+1, n-p-1),$$

so $\{\beta : (\hat\beta-\beta)'X'X(\hat\beta-\beta) \le (p+1)MSE\, F_\alpha(p+1,n-p-1)\}$ is a $100(1-\alpha)\%$ confidence ellipsoid for $\beta$.

**Proof.** $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta-\beta)'X'X(\hat\beta-\beta) = \epsilon' H \epsilon$, and $\epsilon' H\epsilon/\sigma^2 \sim \chi^2(p+1)$. It is independent of $SSE = \epsilon'(I-H)\epsilon$, since $H(I-H)=0$. ∎

Separate 95% intervals for $\beta_0, \ldots, \beta_p$ do not have joint coverage 95%. Two remedies:

**Theorem 8.3 (Bonferroni and Scheffé intervals)**

(a) (Bonferroni) The intervals $\hat\beta_j \pm t_{\alpha/(2(p+1))}(n-p-1)\sqrt{c_{jj}MSE}$, $j = 0, \ldots, p$, cover all $\beta_j$ simultaneously with probability at least $1-\alpha$.

(b) (Scheffé) With probability $1-\alpha$, simultaneously for all $\mathbf{a} \in \mathbb{R}^{p+1}$,

$$\mathbf{a}'\beta \in \mathbf{a}'\hat\beta \pm \sqrt{(p+1)F_\alpha(p+1,n-p-1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}.$$

**Proof.** (a) If $A_j$ is the event that the $j$-th interval covers, $P(A_j^c) = \alpha/(p+1)$, and $P(\bigcap A_j) \ge 1 - \sum P(A_j^c) = 1-\alpha$.

(b) By the Cauchy–Schwarz inequality in the inner product defined by $(X'X)^{-1}$,

$$\max_{\mathbf{a}\neq\mathbf{0}} \frac{\{\mathbf{a}'(\hat\beta-\beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta-\beta)'X'X(\hat\beta-\beta),$$

so the event in (b) is exactly the joint confidence region of Theorem 8.2. ∎

Taking $\mathbf{a} = \mathbf{x} = (1, x_1, \ldots, x_p)'$ in (b) gives a **confidence band** for the whole regression surface, valid for all $\mathbf{x}$ at once. For a few coefficients Bonferroni intervals are shorter; Scheffé intervals pay for covering infinitely many combinations.

## 8.4 Testing equality of regression models

Two populations follow

$$\mathbf{y}_1 = X_1\beta_1 + \epsilon_1, \qquad \mathbf{y}_2 = X_2\beta_2 + \epsilon_2,$$

with $n_1$ and $n_2$ observations, design matrices of size $n_i \times (p+1)$, and independent $N(\mathbf{0}, \sigma^2 I)$ errors with a *common* $\sigma^2$. To test $H_0 : \beta_1 = \beta_2$:

(1) Full model: fit each separately, $SSE(F) = SSE_1 + SSE_2$, with $n - 2(p+1)$ degrees of freedom ($n = n_1 + n_2$).

(2) Reduced model: stack $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2')'$ and $X = (X_1', X_2')'$ and fit one $\beta_0$: $SSE(R)$, with $n-(p+1)$ degrees of freedom.

(3) Under $H_0$,

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-2p-2)} \sim F(p+1, n-2p-2).$$

The reduced model is the full model (with block-diagonal design $\mathrm{diag}(X_1, X_2)$ and parameter $(\beta_1', \beta_2')'$) under the $p+1$ linear restrictions $\beta_1 - \beta_2 = \mathbf{0}$, so this is again a general linear hypothesis. For $k$ groups and $H_0 : \beta_1 = \cdots = \beta_k$, the numerator has $(k-1)(p+1)$ degrees of freedom and the denominator $n - k(p+1)$.

The test assumes equal error variances; with $k = 2$ this can be checked first by $F = MSE_1/MSE_2 \sim F(n_1 - p - 1, n_2 - p - 1)$.

# Chapter 10. Regression Diagnostics

## 10.1 What can go wrong

Fitting $\mathbf{y} = X\beta + \epsilon$, $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, by $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ is justified only if the model is adequate. **Regression diagnostics** look for:

(1) violations of the error assumptions (normality, independence, equal variance), studied through residuals;

(2) a wrong functional form, remedied by transforming $y$ or the $x_j$ (Chapters 5 and 11);

(3) near-singularity of $X'X$ (multicollinearity, Chapter 15);

(4) **high-leverage points**: observations whose $\mathbf{x}_i$ is far from the bulk of the design;

(5) **outliers**: observations whose $y_i$ is far from its fitted value;

(6) **influential observations**: observations whose deletion changes the fit substantially.

Points of types (4)–(6) need not coincide: a high-leverage point can lie on the line, and an outlier at the centre of the design may hardly move the fit.

## 10.2 The hat matrix and leverage

The fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X'X)^{-1}X' = (h_{ij})$, where $h_{ij} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_j$ and $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$ is the $i$-th row of $X$.

**Theorem 10.1 (Properties of the hat matrix)**

(a) $H$ is symmetric and idempotent with $\operatorname{rank} H = \operatorname{tr} H = \sum_i h_{ii} = p + 1$.

(b) $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$; hence $0 \le h_{ii} \le 1$ and $|h_{ij}| \le \frac{1}{2}$ for $i \neq j$.

(c) $HX = X$. If the model has an intercept, $H\mathbf{1} = \mathbf{1}$, i.e. every row of $H$ sums to 1, and $h_{ii} \ge 1/n$.

(d) With an intercept, $h_{ii} = \frac{1}{n} + (\mathbf{x}_i^* - \bar{\mathbf{x}})'(\tilde{X}'\tilde{X})^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$, where $\mathbf{x}_i^* = (x_{i1}, \ldots, x_{ip})'$ and $\tilde{X}$ is the centred $n \times p$ matrix of regressors. For $p = 1$, $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

**Proof.** (a) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$.

(b) The $(i,i)$ entry of $H = HH'$ is $\sum_j h_{ij}^2$. Since it is at least $h_{ii}^2$, $h_{ii} \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. For $j \neq i$, $h_{ij}^2 \le h_{ii} - h_{ii}^2 \le \frac{1}{4}$.

(c) $HX = X$ is direct; the intercept column gives $H\mathbf{1} = \mathbf{1}$. For $h_{ii} \ge 1/n$ see (d).

(d) Reparametrize with centred regressors: the column space of $X$ is spanned by $\mathbf{1}$ and the columns of $\tilde{X}$, which are orthogonal to $\mathbf{1}$. Hence $H = \frac{1}{n}\mathbf{1}\mathbf{1}' + \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}'$, and the diagonal gives the formula, the second term being nonnegative. For $p = 1$, $\tilde{X}'\tilde{X} = S_{xx}$. ∎

Since $\hat{y}_i = \sum_j h_{ij} y_j$, the weight of $y_i$ in its own fitted value is $h_{ii} = \partial \hat{y}_i / \partial y_i$; $h_{ii}$ is the **leverage** of observation $i$. By (d), it grows with the (Mahalanobis) distance of $\mathbf{x}_i^*$ from the centroid $\bar{\mathbf{x}}$: with $M(i) = (\mathbf{x}_i^* - \bar{\mathbf{x}})'\{\tilde{X}'\tilde{X}/(n-1)\}^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$,

$$M(i) = (n-1)\Big(h_{ii} - \frac{1}{n}\Big).$$

The average leverage is $\bar{h} = (p+1)/n$, and a common rule flags $h_{ii} > 2\bar{h}$ (or $3\bar{h}$) as high leverage.

## 10.3 Residuals

**Theorem 10.2**

$\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$. So $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$, $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, and $\operatorname{Corr}(e_i, e_j) = -h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$.

High-leverage points have small residual variance: the fit is pulled towards them. To compare residuals on a common scale:

**Definition (Scaled residuals)**

With $s^2 = MSE = \mathbf{e}'\mathbf{e}/(n-p-1)$ and $s^2_{(i)}$ the $MSE$ computed without observation $i$:

- the **standardized** (internally studentized) residual is $r_i = \dfrac{e_i}{s\sqrt{1-h_{ii}}}$;
- the **studentized** (externally studentized) residual is $r_i^* = \dfrac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$.

$r_i$ is not $t$-distributed, because $e_i$ and $s$ are dependent (a large $|e_i|$ inflates $s$). Deleting observation $i$ removes the dependence. Let $X_{(i)}$, $\mathbf{y}_{(i)}$ denote $X$, $\mathbf{y}$ without row $i$, $\hat\beta_{(i)}$ the corresponding estimator, and $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)}$ the prediction of $y_i$ from the other observations.

**Theorem 10.3 (Deletion formulas)**

(a) $\hat\beta_{(i)} = \hat\beta - \dfrac{(X'X)^{-1}\mathbf{x}_i e_i}{1-h_{ii}}$.

(b) $y_i - \tilde y_i = \dfrac{e_i}{1-h_{ii}}$ (the **deleted** or PRESS residual).

(c) $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - \dfrac{e_i^2}{1-h_{ii}}$.

**Proof.** (a) $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. With $A = X'X$, the Sherman–Morrison formula gives

$$(A - \mathbf{x}_i\mathbf{x}_i')^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}.$$

Multiplying by $X'\mathbf{y} - \mathbf{x}_i y_i$ and using $A^{-1}X'\mathbf{y} = \hat\beta$, $\mathbf{x}_i'\hat\beta = \hat y_i$, $\mathbf{x}_i'A^{-1}\mathbf{x}_i = h_{ii}$:

$$\hat\beta_{(i)} = \hat\beta - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat y_i - h_{ii}y_i)}{1-h_{ii}} = \hat\beta + \frac{A^{-1}\mathbf{x}_i\{\hat y_i - h_{ii}y_i - (1-h_{ii})y_i\}}{1-h_{ii}} = \hat\beta - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)} = \hat y_i - h_{ii}e_i/(1-h_{ii})$, so $y_i - \tilde y_i = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$.

(c) For $j \neq i$, the deleted residual is $y_j - \mathbf{x}_j'\hat\beta_{(i)} = e_j + h_{ji}e_i/(1-h_{ii})$. Summing squares over $j \neq i$ and using $\sum_j e_j h_{ji} = 0$ (since $H\mathbf{e} = \mathbf{0}$) and $\sum_j h_{ji}^2 = h_{ii}$:

$$SSE_{(i)} = \sum_{j\neq i} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j\neq i} h_{ji}e_j + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j\neq i} h_{ji}^2.$$

Here $\sum_{j\neq i} e_j^2 = SSE - e_i^2$, $\sum_{j\neq i} h_{ji}e_j = -h_{ii}e_i$, and $\sum_{j\neq i} h_{ji}^2 = h_{ii} - h_{ii}^2$. So

$$SSE_{(i)} = SSE - e_i^2 - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

■

**Theorem 10.4 (Distribution of the studentized residual)**

(a) $r_i^* = \dfrac{y_i - \tilde{y}_i}{s_{(i)}\sqrt{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i}} \sim t(n-p-2)$.

(b) $r_i^* = r_i\sqrt{\dfrac{n-p-2}{n-p-1-r_i^2}}$, a monotone function of $r_i$.

**Proof.** (a) $y_i$ is independent of $\hat{\beta}_{(i)}$ and $s_{(i)}^2$, which are computed from the other $n-1$ observations; so $y_i - \tilde{y}_i \sim N(0, \sigma^2\{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i\})$, independent of $(n-p-2)s_{(i)}^2/\sigma^2 \sim \chi^2(n-p-2)$. By Sherman–Morrison, $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1 + h_{ii} + h_{ii}^2/(1-h_{ii}) = 1/(1-h_{ii})$, and with Theorem 10.3(b) the ratio equals $e_i/\{s_{(i)}\sqrt{1-h_{ii}}\}$.

(b) Substitute Theorem 10.3(c), $s_{(i)}^2 = s^2(n-p-1-r_i^2)/(n-p-2)$. ■

So an observation is flagged as an outlier when $|r_i^*| > t_{\alpha/2}(n-p-2)$. Since $n$ tests are performed, a Bonferroni level $\alpha/n$ is safer.

## 10.4 Influence measures

An observation is influential if deleting it changes the fitted model a lot. The measures below combine the residual and the leverage.

**Theorem 10.5 (Influence measures)**

(a) **DFFITS**: $\mathrm{DFFITS}_{(i)} = \dfrac{\hat{y}_i - \tilde{y}_{i(i)}}{s_{(i)}\sqrt{h_{ii}}} = \left(\dfrac{h_{ii}}{1-h_{ii}}\right)^{1/2} r_i^*$, where $\tilde{y}_{i(i)} = \mathbf{x}_i'\hat{\beta}_{(i)}$.

(b) **Cook's distance**: $D_{(i)} = \dfrac{(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)})}{(p+1)s^2} = \dfrac{r_i^2}{p+1}\cdot\dfrac{h_{ii}}{1-h_{ii}}$.

(c) **COVRATIO**: $\mathrm{COVRATIO}_{(i)} = \dfrac{\det\{s_{(i)}^2(X_{(i)}'X_{(i)})^{-1}\}}{\det\{s^2(X'X)^{-1}\}} = \dfrac{1}{(1-h_{ii})}\left(\dfrac{n-p-2+r_i^{*2}}{n-p-1}\right)^{-(p+1)}$.

(d) **Andrews–Pregibon**: with $Z = (X, \mathbf{y})$, $\mathrm{AP}_{(i)} = \dfrac{\det(Z_{(i)}'Z_{(i)})}{\det(Z'Z)} = 1 - h_{ii} - \dfrac{e_i^2}{SSE}$.

**Proof.** (a) By Theorem 10.3(a), $\hat{y}_i - \tilde{y}_{i(i)} = \mathbf{x}_i'(\hat{\beta} - \hat{\beta}_{(i)}) = h_{ii}e_i/(1-h_{ii})$; divide by $s_{(i)}\sqrt{h_{ii}}$.

(b) $(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i = \frac{h_{ii}e_i^2}{(1-h_{ii})^2}$; divide by $(p+1)s^2$.

(c) $\det(X_{(i)}'X_{(i)}) = \det(X'X)\,(1-h_{ii})$ by the matrix determinant lemma, so the ratio is $(s_{(i)}^2/s^2)^{p+1}/(1-h_{ii})$. By Theorem 10.3(c), $s_{(i)}^2/s^2 = (n-p-1-r_i^2)/(n-p-2)$, and by Theorem 10.4(b), $n-p-2+r_i^{*2} = (n-p-2)(n-p-1)/(n-p-1-r_i^2)$. Hence $s_{(i)}^2/s^2 = (n-p-1)/(n-p-2+r_i^{*2})$, which gives the formula.

(d) By the determinant lemma, $\det(Z_{(i)}'Z_{(i)})/\det(Z'Z) = 1 - \mathbf{z}_i'(Z'Z)^{-1}\mathbf{z}_i$, where $\mathbf{z}_i' = (\mathbf{x}_i', y_i)$. The last quantity is the leverage of observation $i$ in the regression on $Z$, which by the partitioned-inverse formula equals $h_{ii} + e_i^2/SSE$. ■

Rules of thumb (Belsley, Kuh and Welsch; Cook):

- $|\mathrm{DFFITS}_{(i)}| > 2\sqrt{(p+1)/n}$;

- $D_{(i)} > F_{0.50}(p+1, n-p-1)$ (roughly 1): deleting $i$ moves $\hat{\beta}$ outside the 50% confidence ellipsoid of Theorem 8.2;
- $|\text{COVRATIO}_{(i)} - 1| > 3(p+1)/n$;
- small $\text{AP}_{(i)}$.

Every measure grows with both $|r_i|$ and $h_{ii}$: influence is "outlyingness $\times$ leverage".

**Example 10.1 (Two kinds of unusual point)**

In a classical data set of $n = 21$ children (age at first word $x$, aptitude score $y$), the fit is $\hat{y} = 109.87 - 1.13x$. Observation 18 has $x = 42$, far from the others, with $h_{18,18} = 0.65 > 3\bar{h} = 0.29$: a high-leverage point. Its residual is small ($r^* = -0.85$), yet $D_{(18)} = 0.68$ and $\text{DFFITS}_{(18)} = -1.16$ are the largest: it is influential. Observation 19 has the largest residual, $r^*_{19} = 3.61 > t_{0.025}(18) = 2.10$: an outlier, but with ordinary leverage ($h = 0.053$), so its Cook's distance is only 0.22. The two points call for different actions: check the measurement of the outlier, and assess whether the conclusions depend on the single high-leverage point.

**What to do.** An unusual point is first checked for recording errors. If it is genuine, it may signal a wrong model (a missing variable or nonlinearity); robust regression or reporting fits with and without the point are better than silently deleting it. Influence diagnostics based on deleting one point at a time can miss groups of points that mask each other.

# Chapter 11. Variable Selection

With many candidate regressors we want a **parsimonious** model: enough variables to explain $y$ well and predict accurately, but no more, since every extra variable adds variance to the estimates and predictions. This chapter quantifies the trade-off and describes search methods and criteria.

## 11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where $X_k$ $(n \times (k+1))$ contains the intercept and $k$ regressors and $X_r$ $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

**Theorem 11.1**

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

**Proof.** (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE = Var +bias · bias′.

(b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$.

(c) MSE = Var +bias$^2$ with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

**Theorem 11.2**

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

**Proof.** By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat\beta_k - \operatorname{Var}\tilde\beta_k = \sigma^2 AB^{-1}A' \geq 0$.

(b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat{\beta}_r = \sigma^2 B^{-1}$, so the assumption reads $\sigma^2 B^{-1} \geq \beta_r\beta_r'$. Then $\operatorname{Var}\hat{\beta}_k - \operatorname{MSE}(\tilde{\beta}_k) = A(\sigma^2 B^{-1} - \beta_r\beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat{\beta}) - \operatorname{MSE}(\mathbf{x}_k'\tilde{\beta}_k) = \mathbf{u}'(\sigma^2 B^{-1} - \beta_r\beta_r')\mathbf{u} \geq 0$. ■

In words: dropping variables whose coefficients are small relative to their standard errors (so that $\beta_r\beta_r' \leq \operatorname{Var}\hat{\beta}_r$) reduces the mean squared error of estimation and prediction, even though it introduces bias. Conversely, if the reduced model is true ($\beta_r = \mathbf{0}$), both estimators are unbiased and the reduced one has smaller variance, so including irrelevant variables only costs precision.

## 11.2 Search methods

With $p$ candidate regressors there are $2^p$ subsets (each variable in or out), so exhaustive search becomes infeasible for large $p$ ($2^{50} \approx 10^{15}$).

**All possible regressions (best subset).** Fit all $2^p$ models. For each size $k$, keep the best model by $R_k^2$ (equivalently by $SSE_k$ or $MSE_k$, since these are monotone in each other for fixed $k$), then compare the best models across $k$ with a criterion that penalizes size (Section 11.3).

**Backward elimination.**

(1) Fit the full model.

(2) Compute the partial $F$ statistic of each variable in the model.

(3) Let $F_L$ be the smallest, and $F_C$ the critical value at level $\alpha_{\text{drop}}$.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

**Forward selection.** Start with no regressors. At each stage add the variable that increases $R^2$ the most (equivalently, has the largest partial $F$ given those already in), provided its partial $F$ exceeds the critical value at level $\alpha_{\text{add}}$; otherwise stop.

**Stepwise selection.** Forward selection, but after each addition the variables already in the model are re-examined by partial $F$ tests and removed if they fall below the $\alpha_{\text{drop}}$ threshold (with $\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling). A variable that was useful early can become redundant once others enter.

The methods need not agree, and none guarantees the best subset. The levels used are not true significance levels, because of the repeated testing.

**Example 11.1**

In Hald's classical cement data ($n = 13$, four ingredients $x_1, \ldots, x_4$), $x_2$ and $x_4$ are highly correlated ($x_1+x_2+x_3+x_4 \approx 100$). Backward elimination ($\alpha = 0.05$) removes $x_3$ (partial $F = 0.02$) and then $x_4$ (partial $F = 1.86 < F_{0.05}(1, 9) = 5.12$), ending with $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters $x_4$ first (largest single $R^2 = 0.675$), then $x_1$, and stops because $x_2$ has partial $F = 5.03 < 5.12$, ending with $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise selection with $\alpha_{\text{add}} = 0.10$ enters $x_2$ and then removes $x_4$ (partial $F = 1.86$), ending with $\{x_1, x_2\}$. Because $x_2$ and $x_4$ carry nearly the same information, different routes end at different, almost equally good models.

## 11.3 Selection criteria

For a model with $k$ regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n-k-1)$;

(2) $R_k^2 = 1 - SSE_k/SST$;

(3) adjusted $R^2$: $R_{a,k}^2 = 1 - \dfrac{n-1}{n-k-1}(1-R_k^2) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the $MSE$ of the full model.

$R_k^2$ never decreases when a variable is added, so it can only compare models of the same size. $MSE_k$ can increase when a variable is added (the loss of one degree of freedom may outweigh the decrease in $SSE$), and $R_{a,k}^2$ is a decreasing function of $MSE_k$: maximizing $R_a^2$ is minimizing $MSE_k$.

**Mallows' $C_p$.** We judge the reduced model by the total standardized mean squared error of its fitted values at the $n$ design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}_{ki}'\tilde\beta_k.$$

**Theorem 11.3**

$$\Gamma_k = (k+1) + \frac{\beta_r' X_r'(I-H_k)X_r\beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence $C_k$ estimates $\Gamma_k$, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

**Proof.** By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}_{ki}'(X_k'X_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X_k'X_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}_{ki}'\Big\} = \mathrm{tr}\{(X_k'X_k)^{-1}X_k'X_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta_r'X_r'(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ∎

So we look for models with small $C_k$ and $C_k$ close to $k+1$; a $C_k$ well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.

# Chapter 12. Generalized Least Squares

## 12.1 The general linear model

So far the errors were uncorrelated with equal variance. In time series, clustered or heteroscedastic data this fails. The **general linear regression model** is

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 V),$$

with $V$ a known positive definite matrix, $X$ of size $n \times (p+1)$ and full column rank. Ordinary least squares (OLS) is no longer the BLUE.

Since $V$ is symmetric positive definite, $V = P\Lambda P'$ with $P$ orthogonal, and $T = P\Lambda^{1/2}$ satisfies $V = TT'$. Multiply the model by $T^{-1}$:

$$\mathbf{z} = W\beta + \eta, \qquad \mathbf{z} = T^{-1}\mathbf{y}, \quad W = T^{-1}X, \quad \eta = T^{-1}\epsilon,$$

and $\operatorname{Var}\eta = \sigma^2 T^{-1}V(T^{-1})' = \sigma^2 T^{-1}TT'(T')^{-1} = \sigma^2 I_n$. The transformed model satisfies the usual assumptions.

**Theorem 12.1 (Generalized least squares)**

The BLUE of $\beta$ is the **generalized least squares** (GLS) estimator

$$\hat\beta_G = (W'W)^{-1}W'\mathbf{z} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y},$$

which minimizes $(\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. It satisfies $E\hat\beta_G = \beta$ and $\operatorname{Var}\hat\beta_G = \sigma^2(X'V^{-1}X)^{-1}$.

**Proof.** Apply the Gauss–Markov theorem (Theorem 7.1) to the transformed model; linear unbiased estimators in $\mathbf{z}$ are the same as those in $\mathbf{y}$, since $T$ is invertible. Using $(T^{-1})'T^{-1} = (TT')^{-1} = V^{-1}$, $W'W = X'V^{-1}X$ and $W'\mathbf{z} = X'V^{-1}\mathbf{y}$. Also $\|\mathbf{z} - W\beta\|^2 = (\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. Finally $E\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}X\beta = \beta$ and $\operatorname{Var}\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}(\sigma^2V)V^{-1}X(X'V^{-1}X)^{-1} = \sigma^2(X'V^{-1}X)^{-1}$. ∎

When $V$ is diagonal, $V = \operatorname{diag}(v_i)$, GLS is **weighted least squares** with weights $w_i = 1/v_i$: minimize $\sum w_i(y_i - \mathbf{x}_i'\beta)^2$, giving less weight to noisier observations.

## 12.2 Properties

**OLS under correlated errors.** The OLS estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is still unbiased, but

$$\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1},$$

which is at least $\sigma^2(X'V^{-1}X)^{-1}$ (by Gauss–Markov), with equality when $V = I$ (more generally, when the column space of $X$ is invariant under $V$).

**Residuals and $\sigma^2$.** The GLS residual vector is $\mathbf{e}_G = \mathbf{y} - X\hat\beta_G = \{I - X(X'V^{-1}X)^{-1}X'V^{-1}\}\mathbf{y}$, with

$$E\mathbf{e}_G = \mathbf{0}, \qquad \operatorname{Var}\mathbf{e}_G = \sigma^2\{V - X(X'V^{-1}X)^{-1}X'\}.$$

**Theorem 12.2**

$SSE(\hat\beta_G) = \mathbf{e}_G' V^{-1}\mathbf{e}_G$ satisfies $E\,SSE(\hat\beta_G) = (n-p-1)\sigma^2$, so

$$\tilde\sigma^2 = \frac{(\mathbf{y} - X\hat\beta_G)' V^{-1}(\mathbf{y} - X\hat\beta_G)}{n-p-1}$$

is unbiased for $\sigma^2$, and $\widehat{\mathrm{Var}}(\hat\beta_G) = \tilde\sigma^2 (X'V^{-1}X)^{-1}$.

**Proof.** $SSE(\hat\beta_G)$ is the residual sum of squares of the transformed model, $\|\mathbf{z} - W\hat\beta_G\|^2$, whose expectation is $(n-p-1)\sigma^2$ (Theorem 6.2). Directly: by Theorem 5.4(a),

$$\begin{aligned} E(\mathbf{e}_G' V^{-1}\mathbf{e}_G) &= \sigma^2 \mathrm{tr}[V^{-1}\{V - X(X'V^{-1}X)^{-1}X'\}] \\ &= \sigma^2\{n - \mathrm{tr}((X'V^{-1}X)^{-1}X'V^{-1}X)\} = \sigma^2(n-p-1). \end{aligned}$$

■

By contrast, the OLS estimate $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$ has

$$E\hat\sigma^2 = \frac{\sigma^2}{n-p-1}\left[\mathrm{tr}V - \mathrm{tr}\{(X'X)^{-1}X'VX\}\right],$$

which in general differs from $\sigma^2$, so the usual OLS standard errors $\hat\sigma^2(X'X)^{-1}$ are biased in two ways under correlated errors.

Under normality, all the inference of Chapters 6–8 carries over to the transformed model: for instance $\hat\beta_G \sim N(\beta, \sigma^2(X'V^{-1}X)^{-1})$, $(n-p-1)\tilde\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and $t$ and $F$ tests use $\tilde\sigma^2$.

**Example 12.1 (Averages with unequal precision)**

Suppose $y_i = \mu + \epsilon_i$ where $y_i$ is the average of $m_i$ measurements, so $\mathrm{Var}\,\epsilon_i = \sigma^2/m_i$ and $V = \mathrm{diag}(1/m_i)$. Then $X = \mathbf{1}$ and

$$\hat\mu_G = \frac{\sum m_i y_i}{\sum m_i}, \qquad \mathrm{Var}\,\hat\mu_G = \frac{\sigma^2}{\sum m_i},$$

the grand average of all measurements. The OLS estimate $\bar y$ has variance $\frac{\sigma^2}{n^2}\sum \frac{1}{m_i} \geq \frac{\sigma^2}{\sum m_i}$ by the harmonic–arithmetic mean inequality.

**Example 12.2 (AR(1) errors)**

If $\epsilon_t = \rho\epsilon_{t-1} + u_t$ with $|\rho| < 1$ and $u_t$ i.i.d. $N(0,\sigma_u^2)$, then $\mathrm{Var}\,\epsilon = \frac{\sigma_u^2}{1-\rho^2}V$ with $V_{st} = \rho^{|s-t|}$. The transformation $T^{-1}$ amounts to the familiar quasi-differencing $y_t - \rho y_{t-1}$, $\mathbf{x}_t - \rho\mathbf{x}_{t-1}$ for $t \geq 2$, with the first observation scaled by $\sqrt{1-\rho^2}$. In practice $\rho$ is unknown and is estimated from OLS residuals (feasible GLS).

**Unknown $V$.** $V$ is rarely known. It is modelled with few parameters (equal correlation within clusters, AR(1) in time, variance proportional to a power of a regressor) and estimated from the data, typically from OLS residuals; GLS with the estimate plugged in is **feasible GLS**.

# Chapter 13. Indicator Variables

## 13.1 One indicator variable

Qualitative explanatory variables (sex, treatment, region) enter a regression through **indicator** (dummy) variables taking the values 0 and 1.

**Parallel lines.** Suppose $y$ is the score on a test, $x_1$ the hours of preparation, and $x_2 = 1$ for students who attended a review session and 0 otherwise. The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

gives two response functions with a common slope:

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + \beta_1 x_1, & x_2 = 1. \end{cases}$$

$\beta_1$ is the effect of $x_1$ in both groups, and $\beta_2$ is the vertical distance between the parallel lines: the group effect at any fixed $x_1$. Fitting one model, rather than one line per group, uses all $n$ observations to estimate the common slope and $\sigma^2$.

**Interaction.** If the effect of $x_1$ may differ between groups, add the product term:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i, \qquad E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1, & x_2 = 1. \end{cases}$$

Now $\beta_2$ is the difference in intercepts and $\beta_3$ the difference in slopes. The hypotheses of Chapter 5 become ordinary tests in one model:

- parallel lines: $H_0 : \beta_3 = 0$ (a $t$ test);
- identical lines: $H_0 : \beta_2 = \beta_3 = 0$ (a partial $F$ test with 2 degrees of freedom).

These agree with the separate-fit tests of Section 5.3 because the interaction model with $n_1 + n_2$ observations has exactly the four parameters of the two separate lines and the same residual sum of squares.

**Example 13.1**

In a study of 20 firms, $y$ is the time to adopt an innovation, $x_1$ the firm size and $x_2 = 1$ for stock companies. The fit $\hat{y} = 33.87 - 0.102 x_1 + 8.06 x_2$ says that adoption is faster for larger firms (about 1 month per 10 size units) and that stock companies take about 8 months longer at every size. Adding the interaction gives $\hat{\beta}_3 = -0.0004$ with standard error 0.018, so $t_0 = -0.02$: the parallel-lines model is adequate.

## 13.2 Several categories

A qualitative variable with $q$ categories needs $q - 1$ indicators. With three categories $E_1, E_2, E_3$, define

$$x_2 = \mathbf{1}\{E_1\}, \qquad x_3 = \mathbf{1}\{E_2\},$$

with $E_3$ the **baseline**:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \qquad E(y) = \begin{cases} (\beta_0 + \beta_2) + \beta_1 x_1, & E_1, \\ (\beta_0 + \beta_3) + \beta_1 x_1, & E_2, \\ \beta_0 + \beta_1 x_1, & E_3. \end{cases}$$

So $\beta_2$ and $\beta_3$ compare $E_1$ and $E_2$ with the baseline, and $\beta_2 - \beta_3$ compares $E_1$ with $E_2$.

**The dummy variable trap.** Using $q$ indicators together with an intercept makes the columns of $X$ linearly dependent: the indicators sum to the intercept column $\mathbf{1}$. Then $\operatorname{rank} X <$ number of columns, $X'X$ is singular, and $\beta$ is not identifiable. Either drop one indicator (baseline coding, as above), drop the intercept, or impose a constraint such as $\sum(\text{effects}) = 0$.

**Interactions and several factors.** Separate slopes are obtained with the products $x_1 x_2$ and $x_1 x_3$:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3,$$

so that in $E_1$ the line is $(\beta_0 + \beta_2) + (\beta_1 + \beta_4) x_1$, and so on. Several qualitative factors each get their own set of indicators, and interactions between factors are products of their indicators. A model containing only indicators (no quantitative regressors) is an analysis of variance model written as a regression.

## 13.3 Piecewise linear regression

Sometimes the slope changes at a known point $x_w$ (a threshold or a policy change), while the response stays continuous. With $x_2 = \mathbf{1}\{x_1 > x_w\}$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 (x_{i1} - x_w) x_{i2} + \epsilon_i,$$

so that

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_1 \le x_w, \\ (\beta_0 - \beta_2 x_w) + (\beta_1 + \beta_2) x_1, & x_1 > x_w. \end{cases}$$

The two lines meet at $x_w$, $\beta_2$ is the change in slope, and $H_0 : \beta_2 = 0$ tests for a break. The regressor $(x_1 - x_w)_+ = (x_1 - x_w) x_2$ is a **truncated power basis** function; several knots $x_{w_1} < x_{w_2} < \cdots$ add one such term each, giving a linear spline. If a jump at $x_w$ is also allowed, add $x_2$ itself.

**Example 13.2**

Unit cost $y$ against lot size $x_1$ for eight lots, with a suspected change at $x_w = 500$:

| $x_1$ | 300 | 340 | 400 | 480 | 570 | 650 | 720 | 800 |
|---|---|---|---|---|---|---|---|---|
| $y$ | 475 | 440 | 452 | 377 | 355 | 257 | 249 | 139 |

The fit is $\hat{y} = 589.5 - 0.395 x_1 - 0.389 (x_1 - 500)_+$: the cost falls by 0.40 per unit up to 500 and by $0.40 + 0.39 = 0.78$ per unit beyond.

# Chapter 15. Multicollinearity and Biased Estimation

The least squares estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is the BLUE with covariance $\sigma^2(X'X)^{-1}$. When $X'X$ is nearly singular this covariance is huge, and a slightly biased estimator with much smaller variance can be far more accurate. This chapter studies the problem and two remedies.

Throughout we standardize: $y$ is centred and each regressor is centred and scaled so that $\sum_i x_{ij} = 0$ and $\sum_i x_{ij}^2 = 1$. Then $X$ is $n \times p$ (no intercept column), $X'X$ is the correlation matrix of the regressors, and $X'\mathbf{y}$ is proportional to their correlations with $y$. Coefficients on the original scale are recovered from $\tilde\beta_j = \hat\beta_j\, s_y/s_j$ and $\tilde\beta_0 = \bar y - \sum_j \tilde\beta_j \bar x_j$, where $s_j$ is the scale used for $x_j$.

## 15.1 Multicollinearity

**Multicollinearity** means that the regressors are (nearly) linearly dependent. Exact dependence makes $X$ rank-deficient and $\hat\beta$ undefined; near dependence makes $(X'X)^{-1}$ huge and $\hat\beta$ unstable.

**Example 15.1 (Two regressors)**

With two standardized regressors with correlation $r_{12}$,

$$X'X = \begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}, \qquad \operatorname{Var}\hat\beta = \frac{\sigma^2}{1-r_{12}^2}\begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix},$$

so $\operatorname{Var}\hat\beta_j = \sigma^2/(1-r_{12}^2) \to \infty$ as $|r_{12}| \to 1$, and $\operatorname{Corr}(\hat\beta_1,\hat\beta_2) = -r_{12}$. With $r_{12} = 0.99$ the variances are 50 times those under orthogonal regressors, and confidence intervals $\hat\beta_j \pm t_{\alpha/2}(n-3)\sqrt{MSE/(1-r_{12}^2)}$ are 7 times as wide.

**Eigenvalues.** If $\lambda_1,\dots,\lambda_p$ are the eigenvalues of $X'X$,

$$\sum_j \operatorname{Var}\hat\beta_j = \sigma^2 \operatorname{tr}(X'X)^{-1} = \sigma^2 \sum_j \frac{1}{\lambda_j},$$

so a single small eigenvalue blows up the total variance. The eigenvector of a small eigenvalue shows which combination of regressors is nearly constant.

**Theorem 15.1 (Variance inflation factor)**

Let $R_j^2$ be the coefficient of determination of the regression of $x_j$ on the other regressors. Then $c_{jj}$, the $j$-th diagonal entry of $(X'X)^{-1}$, equals $(1-R_j^2)^{-1}$, so

$$\operatorname{Var}\hat\beta_j = \frac{\sigma^2}{1-R_j^2}.$$

**Proof.** Write $X = (\mathbf{x}_j, X_{-j})$. By the partitioned-inverse formula,

$$c_{jj} = \{\mathbf{x}_j'\mathbf{x}_j - \mathbf{x}_j'X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'\mathbf{x}_j\}^{-1} = \{\mathbf{x}_j'(I-H_{-j})\mathbf{x}_j\}^{-1},$$

which is the inverse of the residual sum of squares of $\mathbf{x}_j$ on $X_{-j}$. Since $\mathbf{x}_j$ is centred with $\mathbf{x}_j'\mathbf{x}_j = 1$, its total sum of squares is 1, so the residual sum of squares is $1 - R_j^2$. ∎

$\mathrm{VIF}_j = (1-R_j^2)^{-1}$ is the **variance inflation factor**: the factor by which $\operatorname{Var}\hat\beta_j$ exceeds its value under orthogonal regressors. $\mathrm{VIF}_j > 10$ (i.e. $R_j^2 > 0.9$) is a common warning sign.

## 15.2 Ridge regression

Measure the accuracy of an estimator $b$ by the total mean squared error $E\|b-\beta\|^2$. For least squares,

$$E\|\hat\beta-\beta\|^2=\operatorname{tr}\operatorname{Var}\hat\beta=\sigma^2\sum_j\frac{1}{\lambda_j}.$$

**Definition (Ridge estimator)**

For $k>0$, $\hat\beta(k)=(X'X+kI_p)^{-1}X'\mathbf{y}$, the minimizer of $\|\mathbf{y}-X\beta\|^2+k\|\beta\|^2$.

Adding $k$ to the diagonal lifts all eigenvalues by $k$, so $X'X+kI$ is well conditioned. If $X'X=I$, then $\hat\beta(k)=\hat\beta/(1+k)$: ridge **shrinks** least squares towards $\mathbf{0}$. In general $\hat\beta(k)=(X'X+kI)^{-1}X'X\hat\beta$, which is biased: $E\hat\beta(k)=(X'X+kI)^{-1}X'X\beta\neq\beta$.

**Canonical form.** Let $X'X=P\Lambda P'$ with $P$ orthogonal and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_p)$. With $Z=XP$ and $\alpha=P'\beta$, the model is $\mathbf{y}=Z\alpha+\epsilon$ with $Z'Z=\Lambda$. Then $\hat\alpha=\Lambda^{-1}Z'\mathbf{y}$, $\operatorname{Var}\hat\alpha=\sigma^2\Lambda^{-1}$, and

$$\hat\alpha(k)=P'\hat\beta(k)=(\Lambda+kI)^{-1}Z'\mathbf{y},\qquad \hat\alpha_j(k)=\frac{\lambda_j}{\lambda_j+k}\hat\alpha_j:$$

each canonical coefficient is shrunk, most strongly in directions with small $\lambda_j$. Since $P$ is orthogonal, $\|\hat\beta(k)-\beta\|=\|\hat\alpha(k)-\alpha\|$.

**Theorem 15.2 (Mean squared error of ridge)**

$$E\|\hat\beta(k)-\beta\|^2=\sigma^2\sum_{j=1}^p\frac{\lambda_j}{(\lambda_j+k)^2}+k^2\sum_{j=1}^p\frac{\alpha_j^2}{(\lambda_j+k)^2}.$$

The first term (total variance) decreases in $k$ and the second (squared bias) increases. There is always a $k>0$ with $E\|\hat\beta(k)-\beta\|^2<E\|\hat\beta-\beta\|^2$.

**Proof.** $\hat\alpha_j(k)$ has mean $\frac{\lambda_j}{\lambda_j+k}\alpha_j$ and variance $\frac{\lambda_j^2}{(\lambda_j+k)^2}\cdot\frac{\sigma^2}{\lambda_j}$; its bias is $-\frac{k}{\lambda_j+k}\alpha_j$. Adding variance and squared bias over $j$ gives the formula. Call it $g(k)$. Then $g(0)=\sigma^2\sum 1/\lambda_j=E\|\hat\beta-\beta\|^2$, and

$$g'(k)=-2\sigma^2\sum_j\frac{\lambda_j}{(\lambda_j+k)^3}+2k\sum_j\frac{\lambda_j\alpha_j^2}{(\lambda_j+k)^3}.$$

At $k=0$, $g'(0)=-2\sigma^2\sum_j\lambda_j^{-2}<0$, so $g(k)<g(0)$ for small $k>0$. ∎

In the orthonormal case $X'X=I$, $g(k)=\frac{p\sigma^2+k^2\|\alpha\|^2}{(1+k)^2}$, minimized at $k=p\sigma^2/\|\beta\|^2$ (set $g'(k)=0$). More generally, a separate constant $k_j$ for each canonical direction, $\hat\alpha_j=\frac{\lambda_j}{\lambda_j+k_j}\hat\alpha_j$ (generalized ridge), has mean squared error $\sum_j\frac{\sigma^2\lambda_j+k_j^2\alpha_j^2}{(\lambda_j+k_j)^2}$, minimized at $k_j=\sigma^2/\alpha_j^2$.

**Choosing $k$.** The optimal $k$ depends on unknown $\sigma^2$ and $\beta$. Practical choices:

- the **ridge trace**: plot $\hat\beta_j(k)$ against $k$ and take the smallest $k$ at which the coefficients stabilize;
- the plug-in rule $\hat k=p\hat\sigma^2/\hat\beta'\hat\beta$ (Hoerl, Kennard and Baldwin), possibly iterated with $\hat\beta(k)$ in place of $\hat\beta$;

- cross-validation, e.g. minimizing $\text{PRESS}(k) = \sum_i \{y_i - \mathbf{x}_i' \hat{\beta}_{(i)}(k)\}^2$. As in Theorem 10.3, with $A(k) = X(X'X + kI)^{-1}X' = (a_{ij})$ and $e_i = y_i - \mathbf{x}_i'\hat{\beta}(k)$,

$$\text{PRESS}(k) = \sum_{i=1}^{n} \Big( \frac{e_i}{1 - a_{ii}} \Big)^2,$$

so no refitting is needed (leave-one-out cross-validation).

## 15.3 Principal component regression

Let $\mathbf{e}_1, \dots, \mathbf{e}_p$ be orthonormal eigenvectors of $X'X$ with eigenvalues $l_1 \geq \cdots \geq l_p > 0$, and $E_r = (\mathbf{e}_1, \dots, \mathbf{e}_r)$. The **principal components** are the columns of $Z_r = XE_r$, and they are orthogonal: $Z_r'Z_r = \text{diag}(l_1, \dots, l_r)$.

**All components ($r = p$).** $X\beta = XE_pE_p'\beta = Z\alpha$ with $\alpha = E_p'\beta$, and

$$\hat{\beta} = E_p\hat{\alpha} = E_p(Z'Z)^{-1}Z'\mathbf{y} = \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}, \qquad \text{Var}\,\hat{\beta} = \sigma^2 \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

which is just least squares.

**Leading components ($r < p$).** Regress $\mathbf{y}$ on $Z_r$ only and transform back:

$$\tilde{\beta}_r = E_r(Z_r'Z_r)^{-1}Z_r'\mathbf{y} = \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}.$$

**Theorem 15.3 (Principal component regression)**

$$E\tilde{\beta}_r = \beta - \sum_{k=r+1}^{p} \mathbf{e}_k\mathbf{e}_k'\beta, \qquad \text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

and

$$E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k} + \sum_{k=r+1}^{p} (\mathbf{e}_k'\beta)^2.$$

**Proof.** Using $X'X = \sum_h l_h\mathbf{e}_h\mathbf{e}_h'$ and orthonormality, $E\tilde{\beta}_r = \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'X\beta = \sum_{k\leq r} \mathbf{e}_k\mathbf{e}_k'\beta$, and $\beta = \sum_{k\leq p} \mathbf{e}_k\mathbf{e}_k'\beta$. Next,

$$\text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k,k'\leq r} \frac{1}{l_k l_{k'}}\mathbf{e}_k\mathbf{e}_k'X'X\mathbf{e}_{k'}\mathbf{e}_{k'}' = \sigma^2 \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'.$$

The trace of the variance plus the squared length of the bias gives the mean squared error. ■

Dropping the components with the smallest $l_k$ removes the largest variance terms $\sigma^2/l_k$ at the cost of the bias $\sum_{k>r}(\mathbf{e}_k'\beta)^2$. This pays off when the small-eigenvalue directions carry little of $\beta$. PCR is a "hard" version of ridge: ridge shrinks every canonical coefficient by $l_k/(l_k + k)$, PCR keeps the first $r$ and sets the rest to 0.

**Choosing $r$.** Common rules: keep components until the cumulative proportion $\sum_{k\leq r} l_k / \sum_k l_k$ (the share of the total variance of the standardized regressors) is large enough; look for an elbow in the scree plot of $l_k$; or choose $r$ by cross-validation. Note that the components are chosen without looking at $y$; partial least squares instead builds components that are correlated with $y$.

# Chapter 16. Logistic Regression

## 16.1 A model for a binary response

When $y$ takes only the values 0 and 1, $E(y \mid x) = P(y = 1 \mid x) =: \pi(x)$ is a probability. The linear model $y = \beta_0 + \beta_1 x + \epsilon$ is unsuitable: fitted values can leave $[0, 1]$, the errors take only two values for each $x$ (so they are not normal), and $\mathrm{Var}(y \mid x) = \pi(x)\{1 - \pi(x)\}$ varies with $x$. Empirically, $\pi(x)$ often follows an S-shaped curve. We therefore model

$$\pi(x) = G(\beta_0 + \beta_1 x)$$

with a cdf $G$, the inverse **link function**. Common choices: the logistic $G(t) = e^t/(1 + e^t)$, the normal cdf $\Phi$ (probit), and the Gumbel $G(t) = \exp(-e^{-t})$. The logistic choice gives the **logistic regression model**

$$\pi(x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}, \qquad \text{equivalently} \qquad \operatorname{logit} \pi(x) = \log \frac{\pi(x)}{1 - \pi(x)} = \beta_0 + \beta_1 x.$$

The curve increases in $x$ if $\beta_1 > 0$ and decreases if $\beta_1 < 0$.

**Odds and odds ratios.** The **odds** of $y = 1$ are $\pi/(1 - \pi) = P(y = 1)/P(y = 0) \in [0, \infty)$, with $\pi = \text{odds}/(1 + \text{odds})$. Under the logistic model

$$\frac{\pi(x)}{1 - \pi(x)} = e^{\beta_0}(e^{\beta_1})^x, \qquad \frac{\pi(x+1)/\{1 - \pi(x+1)\}}{\pi(x)/\{1 - \pi(x)\}} = e^{\beta_1}.$$

So $e^{\beta_1}$ is the **odds ratio** for a unit increase in $x$, the same at every $x$. For a binary $x$ (exposed $= 1$), $e^{\beta_1}$ is the odds ratio between the two groups; for instance $\hat\beta_1 = 2.0$ means the odds of $y = 1$ are $e^2 \approx 7.4$ times higher in the exposed group.

## 16.2 Estimation

With $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})'$ and $\pi_i = \pi(\mathbf{x}_i) = e^{\mathbf{x}_i'\beta}/(1 + e^{\mathbf{x}_i'\beta})$, independent observations $y_i \sim \text{Bernoulli}(\pi_i)$ give the likelihood $L(\beta) = \prod \pi_i^{y_i}(1 - \pi_i)^{1-y_i}$ and

$$l(\beta) = \sum_{i=1}^{n} \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\} = \sum_{i=1}^{n} \{y_i \mathbf{x}_i'\beta - \log(1 + e^{\mathbf{x}_i'\beta})\}.$$

**Theorem 16.1**

With $\pi(\beta) = (\pi_1, \ldots, \pi_n)'$ and $W(\beta) = \mathrm{diag}\{\pi_i(1 - \pi_i)\}$,

$$\frac{\partial l}{\partial \beta} = X'\{\mathbf{y} - \pi(\beta)\}, \qquad \frac{\partial^2 l}{\partial \beta \partial \beta'} = -X'W(\beta)X.$$

The Hessian is negative definite (if $X$ has full column rank), so $l$ is strictly concave and the MLE, when it exists, is the unique root of $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$.

**Proof.** $\frac{d}{dt}\log(1 + e^t) = \frac{e^t}{1+e^t}$ and $\frac{d^2}{dt^2}\log(1 + e^t) = \frac{e^t}{(1+e^t)^2} = G(t)\{1 - G(t)\}$; apply the chain rule with $t_i = \mathbf{x}_i'\beta$. Since $0 < \pi_i < 1$, $\mathbf{a}'X'WX\mathbf{a} = \sum_i \pi_i(1 - \pi_i)(\mathbf{x}_i'\mathbf{a})^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$. ∎

The likelihood equations are nonlinear and are solved by Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \{X'W^{(t)}X\}^{-1}X'\{\mathbf{y} - \pi^{(t)}\} = \{X'W^{(t)}X\}^{-1}X'W^{(t)}\mathbf{z}^{(t)},$$

with the **working response**

$$\mathbf{z}^{(t)} = X\beta^{(t)} + \{W^{(t)}\}^{-1}\{\mathbf{y} - \pi^{(t)}\}.$$

Each step is a weighted least squares regression of $\mathbf{z}^{(t)}$ on $X$ with weights $\pi_i^{(t)}(1-\pi_i^{(t)})$:

$$\beta^{(t+1)} = \arg\min_{\beta} \{\mathbf{z}^{(t)} - X\beta\}' W^{(t)} \{\mathbf{z}^{(t)} - X\beta\},$$

hence the name **iteratively reweighted least squares** (IRLS). A common start is $\beta^{(0)} = \mathbf{0}$. Since the Hessian does not depend on $\mathbf{y}$, Newton–Raphson coincides with Fisher scoring, and $\{X'W(\hat\beta)X\}^{-1}$ estimates the asymptotic covariance of $\hat\beta$. Wald tests and intervals use $\hat\beta_j/\mathrm{se}(\hat\beta_j) \approx N(0,1)$; likelihood ratio tests compare $2\{l(\hat\beta_F) - l(\hat\beta_R)\}$ with a chi-square distribution.

If the two classes can be perfectly separated by a hyperplane in $x$, $l$ increases without bound along a direction and the MLE does not exist.

## 16.3 Two-by-two tables

Binary exposure ($x$) and outcome ($y$) data arise from three designs:

- **cohort** (prospective) studies follow exposed and unexposed groups and record outcomes;
- **case–control** (retrospective) studies sample cases ($y = 1$) and controls ($y = 0$) and record past exposure;
- **cross-sectional** studies sample individuals and record both at one time.

With counts $n_{xy}$:

| | $y=0$ | $y=1$ | total |
|---|---|---|---|
| $x=0$ | $n_{00}$ | $n_{01}$ | $n_{0+}$ |
| $x=1$ | $n_{10}$ | $n_{11}$ | $n_{1+}$ |

The **odds ratio** is $OR = \dfrac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}$ with $\pi_x = P(y=1 \mid x)$, estimated by $\widehat{OR} = \dfrac{n_{11}n_{00}}{n_{10}n_{01}}$.

**Theorem 16.2**

In the logistic model $\mathrm{logit}\, P(y=1 \mid x) = \beta_0 + \beta_1 x$ with binary $x$, the MLEs satisfy

$$\hat\pi_1 = \frac{e^{\hat\beta_0+\hat\beta_1}}{1+e^{\hat\beta_0+\hat\beta_1}} = \frac{n_{11}}{n_{1+}}, \qquad \hat\pi_0 = \frac{e^{\hat\beta_0}}{1+e^{\hat\beta_0}} = \frac{n_{01}}{n_{0+}}, \qquad e^{\hat\beta_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}.$$

**Proof.** The log-likelihood is

$$l = n_{11}(\beta_0+\beta_1) + n_{01}\beta_0 - n_{1+}\log(1+e^{\beta_0+\beta_1}) - n_{0+}\log(1+e^{\beta_0}).$$

The model is saturated (two parameters for two probabilities), so by invariance the MLEs of $\pi_1, \pi_0$ are the sample proportions $n_{11}/n_{1+}$ and $n_{01}/n_{0+}$. Then $e^{\hat\beta_1} = \frac{\hat\pi_1/(1-\hat\pi_1)}{\hat\pi_0/(1-\hat\pi_0)} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ∎

**Why the odds ratio.** In a case–control study the row proportions do not estimate $\pi_x$, because the numbers of cases and controls are fixed by design. But the odds ratio is symmetric in $x$ and $y$:

$$\frac{P(y=1 \mid x=1)P(y=0 \mid x=0)}{P(y=0 \mid x=1)P(y=1 \mid x=0)} = \frac{P(x=1 \mid y=1)P(x=0 \mid y=0)}{P(x=0 \mid y=1)P(x=1 \mid y=0)},$$

since both equal $\frac{p_{11}p_{00}}{p_{10}p_{01}}$ in terms of the joint cell probabilities. So the odds ratio, and hence the slope $\beta_1$ of a logistic regression, can be estimated from case–control data; only the intercept is distorted by the sampling.

**Example 16.1**

In a trial, 140 subjects receive a supplement and 140 a placebo; 17 and 31 catch a cold, respectively. With $x = 1$ for the supplement, $\widehat{OR} = \frac{17 \cdot 109}{123 \cdot 31} = 0.49$: the odds of a cold are about halved, and $\hat{\beta}_1 = \log 0.49 = -0.72$. The standard error is $\sqrt{\frac{1}{17} + \frac{1}{123} + \frac{1}{31} + \frac{1}{109}} = 0.33$, so the Wald statistic is $-2.19$, significant at the 5% level.